International Journal on Document Analysis and Recognition (IJDAR) (2026) 29:3–39
https://doi.org/10.1007/s10032-025-00526-w

SURVEY

Recent advances in text line segmentation and baseline detection in historical document images: a systematic review

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Received: 23 July 2024 / Revised: 23 March 2025 / Accepted: 7 April 2025 / Published online: 7 May 2025
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Abstract
The purpose of this survey is to provide a comprehensive overview of recent advancements in text line segmentation and baseline detection techniques within the analysis of historical document images. Text line extraction is an essential step in the historical documents image analysis pipeline, as its results significantly impact the accuracy of subsequent processes, such as handwritten text recognition (HTR). Through a multi-stage procedure, we carefully selected 49 peer-reviewed studies published since 2019. Based on careful analysis of these studies, we summarize the information of the existing datasets, describe and categorize different methods, and summarize evaluation protocols. In addition, we compare the results of various methods on benchmark datasets. Finally, we highlight the gaps and suggest directions for future research. We believe that this comprehensive survey will be of great assistance to researchers working in the field of historical document image analysis, as it offers critical insights into the latest advancements and developments, providing a foundation for future research.

Keywords Text line extraction · Baseline detection · Review · Survey · Historical documents

1 Introduction and background

One of the important areas of document image analysis is historical document image analysis. In contrast to modern documents, historical documents were produced using means and techniques that are no longer used, which may affect their visual and structural characteristics. Also, unlike printed documents, handwritten historical manuscripts often exhibit fluctuating baselines, irregular inter-line spacing, marginal annotations, multiple text orientations, and the presence of touching and overlapping characters (see Fig. 1). These characteristics, coupled with various forms of degradation, such as faded ink, holes, stains, or bleed-through, make them much more challenging to analyze. For these reasons, historical document image analysis requires designing and developing special methods to extract and analyze written text and other elements.

Figure 2 illustrates a high-level overview of the typical document image processing pipeline. The process begins with image acquisition and enhancement, where scanned document images are preprocessed to improve quality (e.g., noise reduction, binarization). This is followed by layout analysis, which identifies structural components such as columns, text blocks, and marginalia. Text line detection belongs to this stage. Subsequent processing stages include post-processing, handwriting recognition, information extraction, and additional applications.

The purpose of layout analysis is to determine the physical and/or logical structure of the document. Text line extraction is a part of layout analysis, focusing on identifying and localizing individual lines of text within a document image. The terms 'text line detection', 'text line segmentation', and 'text line extraction' are frequently used indistinctly in the literature, albeit they have slightly different meanings. *Text line detection* involves identifying the location and extent of text lines within a document image. *Text line segmentation* refers to the process of separating text into distinct lines. It involves dividing a block of text into individual lines. *Text line extraction* encompasses both text line detection and text line

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Fig. 1 Challenges in historical document layout analysis: irregular spaces, multi-oriented and fluctuating lines (**a**); marginal notes (**a**, **c**); stains (**b**, **d**); touching and connected characters (**a**, **d**); bleed-though (**c**)

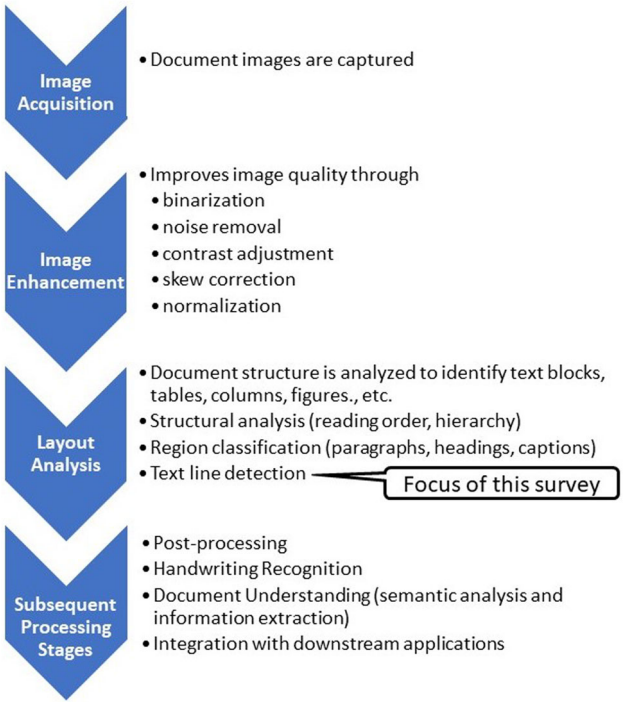


Fig. 2 Typical document processing pipeline. Text line detection is the focus of this survey

segmentation. In this paper, 'text line extraction' is employed as a general term that covers any of these processes.

Text lines can be represented by pixel labels or bounding polygons (Fig. 3 left and middle, respectively). An alternative approach to text line extraction is baseline detection. Baselines are imaginary lines that follow and join the lower parts of characters in a text line (Fig. 3, right).

In this paper, we provide a systematic survey of text line extraction and baseline detection algorithms. Although we focus on algorithms developed for handwritten historical documents, we do not exclude algorithms that apply to both printed and handwritten, or both modern and historical document images. However, we do not survey methods that address general layout analysis. We focus on the last several years, namely since the beginning of 2019, because we aim to review recent trends and advances related to text line extraction and baseline detection in historical documents.

Earlier surveys – most notably the work by Likforman-Sulem et al. [62], published in 2007 as the first survey on text line segmentation in historical documents, and the more recent domain-specific overview by Djaghbellou et al. [21] focusing on Arabic-script manuscripts – have provided valuable overviews of the field. In contrast, this paper adopts a *systematic literature review (SLR) methodology* [55, 83], and to the best of our knowledge, it is the first to apply such a methodology to synthesize and evaluate recent research



Fig. 3 Text line representations: pixels representing the text line are marked by blue (left), bounding box (middle), and baseline (right)

(from 2019 to 2024 including) on text line extraction and baseline detection methods.

The current survey brings the following major contributions to the research community:

- We summarize and categorize recent text line extraction and baseline detection techniques and provide a hierarchical view of the applied methodologies.
- We identify and discuss different performance evaluation protocols.
- We summarize existing handwritten and printed historical datasets that provide annotation on the text line level.
- We summarize and compare the results of different methods on benchmark datasets.
- Finally, we identify research gaps and suggest future research directions.

The results of this survey will be beneficial to both experienced researchers and beginners in the field of historical document image analysis. We believe that this critical analysis will help to understand the current state-of-the-art techniques, identify gaps in the research, and develop novel approaches to enhance the accuracy and efficiency of document analysis algorithms.

2 Research questions

We formulated the following research questions:

RQ_1: **Research Demographic Information**. This question aims to explore the distribution of publication venues and author geolocations across state-of-the-art studies. This is a mostly quantitative analysis that provides an overview of the research groups working on text line extraction and baseline detection methods. This type of information can help in the search for collaboration partners and the locations of the relevant research centers.

RQ_2: **Research Relevance**. This question aims to explore the topic's popularity and its dynamics over the last several years (since the beginning of 2019) by determining the numeric distribution of publications each year.

RQ_3: **Benchmark Datasets**. The purpose of this question is to provide an overview of existing datasets that have been used in various studies. The datasets' description includes extracting the relevant statistics, such as the number of samples, the official split into training, validation, and test sets, the main language in which documents are written, the link to the download of the dataset, etc.

RQ_4: **Methods**. Here, we wanted to explore which methods were used for text line extraction during the past six years and how those methods evolved over time. We also provide a hierarchical view of the applied methods.

RQ_5: **Performance Evaluation Protocols**. The aim of this question is to summarize the current state-of-the-art protocols for the evaluation of baseline detection and text line extraction methods. Having standardized evaluation protocols ensures that the results of different studies can be accurately compared and validated, leading to advancements in the field

RQ_6: **Comparison of the Results**. The aim of this question is to compare and analyze the results of different methods on benchmark datasets. By comparing and analyzing the results of different methods on benchmark datasets, researchers can identify the most effective approaches and potentially improve the accuracy and efficiency of future models or algorithms.

RQ_7: **Available Software Resources**. This question aims to collect information regarding the existence of any publicly available prototypical implementations of the approaches introduced in the retrieved papers.

3 Methodology

We follow the systematic literature review (SLR) procedure as described by Kitchenham [55] and Rammal [83]. The SLR defines three main phases:

1. *Planning the Review.* During this phase, the review protocol is developed, which includes the definition of the research questions the review aims to answer, the strategy for literature search and bibliography management, study selection criteria, and data extraction strategy.
2. *Conducting the Review*. Conducting the review includes documentation of the search, selection of primary studies, data extraction, and data analysis and synthesis. This stage also includes refining the initial protocol.
3. *Reporting the Review.* In this phase, the findings and results are reported.

3.1 Selection of primary studies

For the selection of primary studies, we performed searches within the IEEE Xplore®, ACM Digital Library, and Elsevier ScienceDirect®scientific repositories. Additionally, we extended our search to include Google Scholar. To gain insights into recent developments within the researched area, we limited our search to the studies published since 2019.

The search criteria we applied focused on identifying studies whose titles matched the following regular expression: (S_1 **OR** S_2) **AND** (S_3 **OR** S_4 **OR** S_5) **AND** S_6, where

S_1: **text line**;
S_2: **baseline**;
S_3: **extract*** (extraction OR extracting OR extract);
S_4: **segment*** (segmentation OR segmenting OR segment);
S_5: **detect*** (detection OR detecting OR detect);
S_6: **historical documents**.

Using this search query, we retrieved a large number of studies that focused mainly on text line extraction or baseline detection in handwritten documents. Then, we applied a combined backward and forward snowballing strategy to retrieve relevant studies whose titles do not match the search query.

To ensure the inclusion of only high-quality, peer-reviewed papers, we excluded publications in journals categorized as below Q1-Q2 and conferences with a core rank lower than A or B. Then, we conducted a thorough review of all retrieved studies, eliminating those unrelated to the field of historical document analysis. Following this rigorous selection process, we compiled a list of 49 primary studies that met all our search criteria, with 20 papers published in journals and 29 in conferences.

3.2 Information extraction

To address the stated research questions, we thoroughly analyzed selected publications and extracted necessary information. To facilitate information extraction, we designed the template outlined in Table 1.

4 Findings and analysis

In this section, we report the findings of each research question formulated in Sect. 2. We also discuss and analyze the results.

4.1 Research demographic information

For each source, we looked at the affiliations of all authors to get a sense of the active research organizations and their geographic distribution. After extracting this information, we found that the selected papers were written by researchers from 23 different countries, which are presented in Fig. 4. France leads with eleven papers, followed by China with nine papers, India with eight papers, Israel with six papers, Switzerland with five works, Germany with four papers, and other countries with one or two papers each.

12 papers are written by groups of researchers from two or more countries, demonstrating international collaborations.

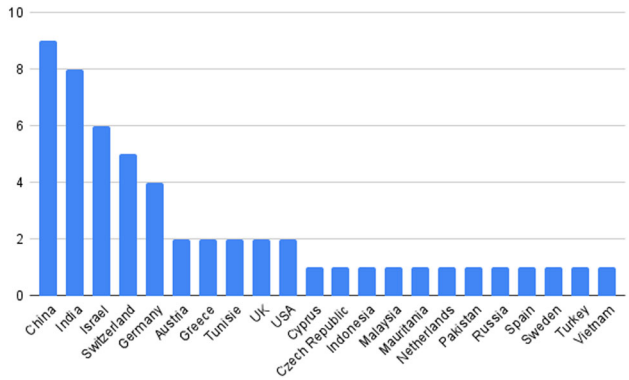


Fig. 4 Demographic information of the research groups according to the affiliations of all authors

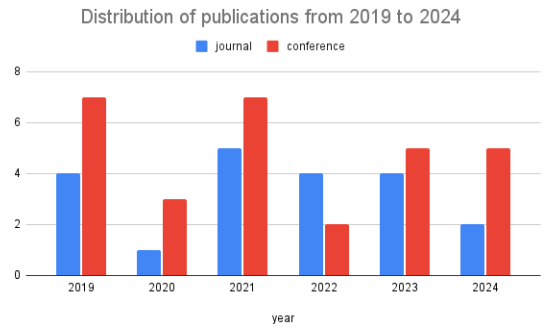


Fig. 5 Number of publications per year

4.2 Research relevance

We explored the dynamics of research interest in text line extraction from 2019 to 2024, including. Figure 5 demonstrates the number of publications per year, categorized as conferences and journals. As can be seen, conference publications dominated this field over the past six years. However, journal publications dominated in 2022 and were almost equal to the number of conference publications in 2023. We also can observe that, despite reaching its peak in 2021, this topic is still active and quite popular.

Table 1 The template for information extraction. We associate each item with the corresponding research question

Title (RQ_1)	Article title
Authors (RQ_1)	The names of all authors
Affiliation (RQ_1)	Authors affiliations (Computer Science/Engineering/Social Sciences) and countries
Year (RQ_2)	The publication year
Publication Type (RQ_2)	Journal/Conference
Rank (RQ_2)	Journal/Conference rank
Dataset(s) (RQ_3)	Utilized dataset(s)
Task(s) (RQ_4)	Text line segmentation or baseline detection
Method (RQ_4)	Applied classifier (supervised/unsupervised, traditional ML/deep-learning)
Evaluation Protocol (RQ_5)	Experimental settings and evaluation metrics
Results (RQ_6)	Reported results
Available Resources (RQ_7)	Link to open source code/available application

4.3 Datasets

Datasets are crucial for the research community, as they allow researchers to validate their models and compare their results with others in the field, fostering collaboration and advancement in the domain. Recently, Nikolaidou et al. [73] published a survey of historical image datasets, categorized into three tasks: document classification, layout structure, and content analysis. While their work provides a valuable foundation, our survey specifically focuses on datasets that offer annotations for text line extraction or baseline detection. Although some of the datasets we summarize were also covered in [73], our review extends beyond the existing literature by including additional datasets and providing further detailed information.

After a thorough analysis of the collected studies, we identified 47 datasets that have annotations on the text line level and were utilized for the task of a text line or baseline detection[1]; 25 of these datasets are available for download. We summarize all the datasets in Table 2. For each dataset, we collected the following information: official name, description, type of the documents (handwritten or printed), primary language, number of samples, number of writers, the annotation level (text lines (TL) or baseline (BL)), ground truth format, the official split into training, validation and test sets, the download link,[2] and references to the studies that utilized the dataset in their experiments.

Overall, the datasets cover 25 different languages (see Fig. 6). Latin and Arabic are the most represented languages, with 12 and 6 datasets, respectively. Additional languages that were explored in different datasets are Chinese (5 datasets), German (4 datasets), French (4 datasets), Greek (3 datasets), Dutch (2 datasets), Tibetan (2 datasets), Balinese (2 datasets), and English, Persian, Hebrew, Khmer, Sudanese, Russian, Cham, Sanskrit, Spanish, Bangla, Devanagari script, Czech, Malayalam, Javanese, Hindi, Telugu explored in one dataset each. Table 3 and Fig. 6 summarize the languages represented by different datasets. For each language, we list the corresponding datasets.

The most extensively utilized datasets are ICDAR 2017 cBAD/READ-BAD (14 studies), DIVA-HisDB (12 studies), and ICDAR 2019 cBAD (8 studies). We overview these datasets in Sects. 4.3.1, 4.3.2, and 4.3.3.

4.3.1 DIVA-HisDB

The DIVA-HisDB dataset was introduced in [89]. The dataset consists of 150 annotated images from three medieval manuscripts: CSG18, CSG863, and CB55, with 50 pages from each manuscript. The CSG18 and CSG863 manuscripts are dated to the 11th century and written in Latin using a Carolingian minuscule script. There is no information regarding the number of writers in these two manuscripts. The CB55 manuscript is dated to the 14th century. It is written in Italian chancery script and contains marginal glosses in Latin. The CB55 manuscript was written by a single author.

The DIVA-HisDB dataset was used in the ICDAR2017 Competition on Layout Analysis for Challenging Medieval Manuscripts [90] for baseline detection (Task-2) and text line segmentation (Task-3). The train, validation, and test sets consist of 30, 10, and 10 pages per manuscript, respectively. The ground truth for the text line segmentation is given in PAGE XML [78] format. The ground truth for the baseline detection was created from the tight polygons of text lines [90]. We identified 14 recent studies that utilized

[1] While many of these datasets include additional types of annotations, in this study we specify only the annotation data related to text line and baseline detection.

[2] For a few datasets [10, 56, 71], the links published in the original papers are no longer active. In such cases, we specified that the dataset is not available for download.

Table 2 Overview of historical document image datasets utilized for text line (TL) and/or baseline (BL) detection with information (if available) about documents types - handwritten(H)/printed(P), language, no. of writers, the annotation level (baseline or text line), ground-truth format (GT), the dataset split, link for download, competition, studies

Dataset	Description	Type	Language	#imgs	#writers	Annotation	GT format	train/val/test	Availability	Studies	Competition
ANDAR-TL: ICDAR2015 [71]	The database was developed by Ancestry.com and consists of about 1300 images drawn from four collections from the 18th and 19th centuries	H	N/A	1300	N/A	origin point detection of text line	MDAT	700/-/600	N/A	[38]	ICDAR 2015 Competition on Text Line Detection
DIVA-HisDB: ICDAR2017 [89]	Consists of three manuscripts: CSG18 and CSG863 dated to 11th century, and CB55 dated to 14th century	H	CSG18 and CSG863: Latin, CB55: Italian	150, 50 from each manuscript	unknown for CSG18 and CSG863, single writer in CB55	BL & TL	pixel-label images and PAGE XML	30/10/10 for each manuscript	https://diuf.unifr.ch/main/hisdoc/diva-hisdb.html	[3, 6, 7, 12, 13, 23, 30, 38, 45, 50, 65, 67, 72]	ICDAR2017 competition on layout analysis for challenging medieval manuscripts
cBad: ICDAR2017 (READ-BAD) [37]	Images collected from 9 different archives, written between 1470 and 1930	H	Latin	2035	N/A	BL & TL	PAGE XML	track A: 216/-/539, track B: 270/-/1011	https://zenodo.org/records/835441	[4, 6, 12, 13, 25, 30, 38, 40, 46, 65–69]	cBAD: ICDAR2017 Competition on Baseline Detection
cBAD: ICDAR2019 [19]	Images sourced from seven European archives, showcasing diverse layouts	both H & P	Latin	3021	N/A	BL & TL	PAGE XML	755/755/1511	https://zenodo.org/records/3568023	[4, 13, 19, 46, 52, 56, 69, 98]	cBAD: ICDAR2019 Competition on Baseline Detection
ICFHR18 RASM2018 [16]	Documents selected from the British Library, 8th-19th	H	Arabic	100	TL	N/A	PAGE XML	15/-/85	https://primaresearch.org/RASM2018/	[66, 67]	Competition on Recognition of Historical Arabic Scientific Manuscripts - RASM2018

Table 2 continued

Dataset	Description	Type	Language	#imgs	#writers	Annotation	GT format	train/val/test	Availability	Studies	Competition
ICFHR18 Asian Palm Leaf [51]	Three different corpora of palm leaf manuscripts written in three different scripts and languages, collected from various locations in Indonesia and Cambodia	H	Balinese, Khmer, and Sudanese	N/A	Balinese: 95, Khmer: 250, Sudanese: 61	TL	BMP	Balinese: 47/-/49, Khmer: 50/-/200, Sudanese: 31/-/30	http://amadi.univ-lr.fr/ICFHR2018_Contest/index.ph0	[1, 77, 96]	ICFHR 2018 Competition On Document Image Analysis Tasks for Southeast Asian Palm Leaf Manuscripts
ICDAR19 RASM2019 [81]	A selection of historical Arabic scientific manuscripts digitised through the British Library, 10th-19th century	H	Arabic	120	N/A	TL	PAGE XML	20/-/100	https://www.primaresearch.org/RASM2019/		RASM2019 ICDAR2019 Competition on Recognition of Historical Arabic Scientific Manuscripts
ICDAR19 HDRC-Chinese [85]	historical database of Chinese family records, written in Chinese traditional Han script	H	Chinese	1172	N/A	TL	PAGE XML	N/A	https://tc11.cvc.uab.es/datasets/ICDAR2019HDRC_1	[44, 47, 60, 85, 88]	ICDAR 2019 Historical Document Reading Challenge on Large Structured Chinese Family Records (ICDAR2019HDRC)
IAM-HistDB [28]	consists of three datasets: Saint Gall, Parzival, and George Washington (GW20)	H	Saint Gall: Latin, Parzival: German, GW20: English	Saint Gall: 60, Parzival: 47, GW20: 20	Saint Gall: single writer, Parzival: 3 writers, GW20: single writer	TL	SVG	N/A	https://diuf.unifr.ch/main/hisdoc/iam-histdb.html	[70]	
BADAM [53]	Documents from a variety of domains and time periods	H	Arabic, Persian	400	N/A	BL	PAGE-XML & bitmasks	320/-/80	https://doi.org/10.5281/zenodo.3274428	[31, 53]	

Table 2 continued

Dataset	Description	Type	Language	#imgs	#writers	Annotation	GT format	train/val/test	Availability	Studies	Competition
HORAE [11]	Dataset of annotated pages from books of hours, 13th -16th	H	Latin	557	N/A	TL	PAGE XML	510/17/30	https://github.com/oriflamms/HORAE/tree/master/Doc_Analysis/Layout_annotation/HORAE_LS	[11–13]	
Pinkas [58]	contains records of Jewish communities in Europe in the early modern period, 1500–1800	H	Hebrew	30	N/A	TL	PAGE XML	24/-/6	https://zenodo.org/records/3569694#.Xe_WdOgzaUk	[4, 23]	
Digital Peter [80]	Peter the Great's manuscripts written between 1709 and 1713	H	Russian	662	single writer	TL	COCO	595/-/67	https://github.com/MarkPotanin/DigitalPeter	[45, 80]	Digital Peter: recognition of Peter the Great's manuscripts [79]
ScribbleLens [22]	Early Modern Dutch documents such as ship journals and daily logbooks produced between the 16th and 18th century	H	Dutch	1000	85	TL	TXT	160/40/800	https://www.openslr.org/84/	[13]	
CHAMDoc [72]	Images obtained from the stamping of the inscriptions, 6th-15th centuries	inscription images	Cham, Sanskrit	26	N/A	TL	mask PNG	N/A	http://l3i-share.univ-lr.fr/2022CHAMDoc/CHAMDoc.html	[72]	
VML-AHTE [24]	Dataset with crowdy diacritics, and touching and overlapping characters	H	Arabic	N/A	30	TL	PAGE XML, pixel labels, Diva pixel labels, bounding polygons	20/-/10	https://beratkurar.github.io/data/ahte_dataset.zip	[7, 23, 45, 59, 75]	

Table 2 continued

Dataset	Description	Type	Language	#imgs	#writers	Annotation	GT format	train/val/test	Availability	Studies	Competition
VML-MOC [5]	Documents with multiply oriented and curved text lines	H	Arabic	N/A	30	TL	raw pixel labeling, DIVA pixel labeling, PAGE XML	20/-/10	https://zenodo.org/records/3559101	[5, 75]	
MTHv2 [64]	Tripitaka Koreana in Han (TKH) and the extended Multiple Tripitaka in Han (MTH) datasets	N/A	Chinese	3199	N/A	TL	N/A	2399/-/800	https://github.com/HCIILAB/MTHv2_Datasets_Release	[32, 44, 47, 60, 64]	
SCUT-CAB [15]	Chinese ancient books containing different binding forms, fonts, and preservation conditions	N/A	Chinese	4000	N/A	TL	JSON	3200/-/800	https://github.com/HCIILAB/SCUT-CAB_Dataset_Release	[60]	
GTLD-small [49]	Consist of three collections (1) ShakeIT of Greek polytonic documents from about 140 books of Greek translations of Shakespeare's plays published between 1842–1950, (2) PIOP documents from the Historical Archives of the Piraeus Bank Group Cultural Foundation, early 20th century (3) Tobacco-3482-small	both P & H	Greek	PIOP: 950, ShakeIT: 357, Tobacco-3482-small: 355	N/A	TL	TXT	PIOP: 672/90/188, ShakeIT: 264/27/66, Tobacco-3482: 240/30/65	https://doi.org/10.5281/zenodo.8020403	[49]	

Table 2 continued

Dataset	Description	Type	Language	#imgs	#writers	Annotation	GT format	train/val/test	Availability	Studies	Competition
Bozen [93]	The pages are extracted from documents of the Ratsprotokolle collection written between 1470 and 1805. This dataset is part of the READ project	H	German	450	N/A	TL	PAGE XML	350/50/50	https://zenodo.org/records/218236	[13, 38, 45]	
PERO [56]	Documents range from handwritten letters to historically printed books and newspapers and contain various languages. Part of the PERO dataset was collected from cBAD, IMPACT, and BADAM datasets	both P & H	multiple languages	683	N/A	BL	N/A	456/-/227	N/A	[56]	
Balsac [97]	Birth, marriage and death records extracted from 74 Quebecois registers	H	French	913	N/A	TL	N/A	730/92/91	N/A	[12, 13]	
ANT [65]	Documents from Tunisian national archives, composed of two sub-datasets of handwritten document images: ANT (Arabic) and ANT (Latin), 17th century	H	Arabic, Latin	Arabic: 180, Latin: 116	N/A	TL	PAGE XML	N/A	N/A	[65–67]	
Hattem [86]	Pages from the C5 Hattem Manuscript, 15th century	H	Dutch	40	single writer	TL & BL	N/A	N/A	N/A	[76]	
No official name [43]	Images selected from the Beijing woodblock edition of the "Kanjur", 1683	H	Tibetan	322	N/A	TL & BL	pixel labels	N/A	N/A	[43]	

Table 2 continued

Dataset	Description	Type	Language	#imgs	#writers	Annotation	GT format	train/val/test	Availability	Studies	Competition
ISIDDI [10]	Contains degraded Bengali pages with different fonts and formats, scanned from 15 old Bengali printed books	P	535	Bengali	–	TL	Unicode	323/130/82	N/A	[57, 70]	
No official name [48]	Different manuscripts and scriptures in Devanagari script of varying fields from 15th to 18th century originated from central India	H	Devanagari script	19889	N/A	TL	N/A	13996/-/5893	N/A	[48]	
HOME-NACR [14]	Charters from the Archives of the Bohemian Crown and archives of monasteries, 1145–1491	H	German, Latin and Czech	496	N/A	TL	N/A	398/49/49	N/A	[13]	
HOME-Alcar [92]	Cartularies produced between the 12th and 14th centuries	H	Latin	55	N/A	TL	PAGE XML	N/A	https://doi.org/10.5281/zenodo.5600884	[13, 30]	
NMU-THDTL [61]	The dataset based on the representative pages of the historical Tibetan book, the "Kanjur", and images synthetically generated images from them	H	Tibetan	200	–	TL & BL	N/A	N/A	https://pan.baidu.com/s/1sgScgacEcJzHDDXZHjWktQ	[61]	
Alcaraz [76]	Documents from the Spanish Inquisition process against Pedro Ruiz de Alcaraz	H	Spanish	44	single writer	TL & BL	N/A	N/A	N/A	[76]	
CHDAC [47]	A complex rotated and warped Chinese historical document analysis dataset	N/A	Chinese	3000	N/A	TL	N/A	2000/-/1000	N/A	[44, 47]	

Table 2 continued

Dataset	Description	Type	Language	#imgs	#writers	Annotation	GT format	train/val/test	Availability	Studies	Competition
AN-Index [13]	Document images of digitized finding aid from the French National Archives	N/A	French	34	N/A	TL	N/A	19/3/12	N/A	[13]	
BNPP [13]	The dataset was provided by the historical Archives of the BNP Paribas bank, written in French between the 19th and 20th century	N/A	French	12	N/A	TL	N/A	7/2/3	N/A	[13]	
Indiscapes2 [82]	Indic manuscripts written in a variety of Indic languages	H	multiple	1275	N/A	TL	image masks	748/-/258	https://ihdia.iiit.ac.in/	[1, 96]	
KgathaM [96]	A palm leaf manuscript containing verses from a poem	H	Malayalam	392	N/A	TL	binary masks	313/-/79	N/A	[1, 96]	
PARES [8]	Handwritten transcriptions of index tables from about 1670 A.D. to 1862 A.D. Tables were copied during the 1960s and 1970s	H	French	250	a few different writers	TL	image masks	80%/10%/10%	https://doi.org/10.5281/zenodo.8337504	[9]	
DRoSB dataset [30]	French documents dating from the 18th century that record the decisions of the States of Burgundy, which, under the Old Regime, governed the province together with the royal power	H	French	240	N/A	polygonal masks	N/A	N/A	N/A	[30]	
CamStd [29]	Images from two historical works from 1564 and	P	Latin	80	–	TL	PAGE XML	N/A	N/A	[29]	

Table 2 continued

Dataset	Description	Type	Language	#imgs	#writers	Annotation	GT format	train/val/test	Availability	Studies	Competition
CamMarg [29]	Historical print from 1595	P	Latin	195	–	TL	PAGE XML	N/A	N/A	[29]	
GW5051 [29]	A print from 1497	P	German	142	–	TL	PAGE XML	N/A	N/A	[29]	
GW5056 [29]	A print from 1497	P	Latin	150	–	TL	PAGE XML	N/A	N/A	[29]	
CamManual [29]	Composed of three historical prints from 1523, 1532, and 1611	P	Greek, Latin	N/A	–	TL	PAGE XML	N/A	N/A	[29]	
GBACHD [100]	woodblock-printed (engraved) texts from various classical books across different dynasties, each with distinct engraved writing styles with complex layouts and font; span different time periods and publishers	H	Chinese	2000	–	TL	JSON	1400/-/600	https://iacc.pazhoulab-huangpu.com/dataset/41.html#:~:text=		
No official name [95]	40 ancient Egyptian papyri in cursive hieratic script coming from the workers' village at Deir el-Medina, Egypt, today housed in the Museo Egizio in Turin	H	Greek	82	–	TL & BL	VOC	80/20/- ratio	N/A	[95]	
No official name [1]	Three palm leaf manuscripts, VM, UB, SM	H	Balinese, Javanese, Hindi, Telugu	VM - 60, UB - 30, SM - 30	multiple	TL	VM -/-/60, UB -/-/30, SM -/-/30	N/A	[1]		

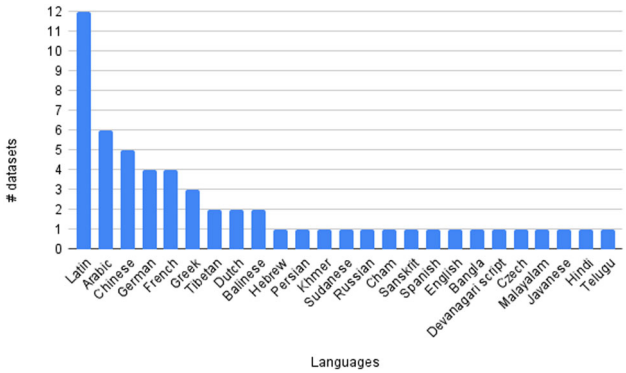


Fig. 6 Number of datasets for each studied language

DIVA-HisDB in their studies. These studies are listed in the "studies" column of Table 2.

4.3.2 READ-BAD

The READ-BAD dataset was introduced in [37] and was utilized in the cBAD ICDAR2017 Competition on Baseline Detection [18]. The dataset consists of 2036 document page images that were collected from nine different archives and written in a wide span of time from 1470 to 1930. In the cBAD 2017 competition, the documents were split into two subsets: documents with a simple layout (track A) and documents with a complex layout (track B). The former category comprises pages featuring simple layouts, whereas the latter encompasses pages with complex layouts containing full-page tables, multi-column text, and rotated text lines. Each subset is split into training and test sets: 216/539 and 270/1011 for tracks A and B, respectively. The READ-BAD dataset provides annotation for both BL and TL tasks.

4.3.3 ICDAR2019 cBAD competition dataset

The cBAD 2019 competition [19] – a successor of the cBAD 2017 competition – also focused on baseline detection. The cBAD 2019 competition introduced a new dataset comprising 3021 page images compiled from seven European archives and showcasing diverse layouts, such as drawings, medieval handwriting, engraving, and historical prints. The dataset encompasses both handwritten and printed images of Latin text in various languages. The dataset is divided into a training set with 755 images, an evaluation set with 755 images, and a test set with 1511 images. Ground truth annotations are provided in PAGE-XML format.

4.4 Methods applied for text line segmentation

We explored the methods applied since the beginning of 2019 for text line segmentation. Figure 7 depicts the hierarchical view of all surveyed methods, while Fig. 8 demonstrates the evolution of the applied methods through time.

Generally speaking, all methods can be organized into the following categories: pure unsupervised analysis, pure supervised analysis, and hybrid methods, which combine supervised and unsupervised stages.

All works that employed a strictly supervised approach did it with deep neural networks (NNs), therefore, this category is not split into additional categories in our hierarchy. Hybrid approaches apply various methods, which we classified into traditional machine learning supervised methods, deep learning, and clustering. Only one work of Pastor [76] applied the traditional ML method—forest of Extremely Randomized Trees—while most supervised methods were again deep NN-based.

There has been an obvious evolution from simple traditional models to deep NNs since the period of our study began. However, every year, almost half of the works still combine supervised classification with unsupervised heuristics.

We put this discussion at the beginning of each subsection, describing supervised and unsupervised approaches, respectively. The conclusions made from this analysis may help researchers understand the advantages and disadvantages of the proposed methods and help them choose the correct approach.

Because the number of studies varied from year to year, we normalized the numbers according to the total number of models reported per year. However, because some papers reported on several models (for example, competition reports), we decided to count every system in our statistics and not just papers (i.e., we provide a percentage of usage for each model per year).

The most popular models in the period that we studied were deep models. Deep models were applied in works from both supervised and hybrid categories. CNNs were mostly applied as end-to-end models that contain a classification layer, and very rarely (we found it in only one work) as feature extractors. Figure 7 depicts the hierarchical view of all the methods applied for the text line detection task since the beginning of 2019. Each leaf of this tree contains the number of works that applied an approach of interest and the section number where these works are discussed. A brief description of the applied approach is also provided in the corresponding section.

4.4.1 Unsupervised analysis

The main advantage of unsupervised analysis is that it does not require any annotated data. Additionally, in most cases, the results of unsupervised learning are easily interpretable, because these methods are based on intuitive heuristic assumptions that can be put into deterministic rules. How-

Table 3 Languages covered by the datasets

Language	Datasets
Latin	DIVA-HisDB:ICDAR2017 [89], IAM-HistDB (SaintGall) [28], cBad:ICDAR2017 [37], cBAD:ICDAR2019 [19], HORAE [11], ANT [65], HOME-NACR [14], HOME-Alcar [92], CamStd [29], CamMarg [29], GW5056 [29], CamManual [29]
Arabic	BADAM [53], ICFHR18 RASM2018 [16], ICDAR19 RASM2019 [81], VML-AHTE [24], VML-MOC [5], ANT [65]
Chinese	ICDAR19 HDRC-Chinese DB [85], MTHv2 [64], SCUT-CAB [15], CHDAC [47], GBACHD [100]
German	IAM-HistDB (Parzival) [28], Bozen [93], HOME-NACR [14],GW5051 [29]
French	Balsac [97], AN-Index [13], BNPP [13], PARES [8],DRoSB [30]
Dutch	ScribbleLens [22], Hattem [86]
Tibetan	NMU-THDTL [61], No official name [43]
English	IAM-HistDB (GW20) [28]
Persian	BADAM [53]
Hebrew	Pinkas [58]
Balinese	ICFHR18 Asian Palm Leaf [51], No offical name [1]
Khmer	ICFHR18 Asian Palm Leaf [51]
Sudanese	ICFHR18 Asian Palm Leaf [51]
Russian	Digital Peter [80]
Cham	CHAMDoc [72]
Sanskrit	CHAMDoc [72]
Spanish	Alcaraz [76]
Greek	GTLD-small [49], CamManual [29], No offical name [95]
Bangla	ISIDDI [10]
Devanagari script	No offical name [48]
Czech	HOME-NACR [14]
Malayalam	KgathaM [96]
Mutiple Indic languages	Indiscapes2 [82], No official name [1]

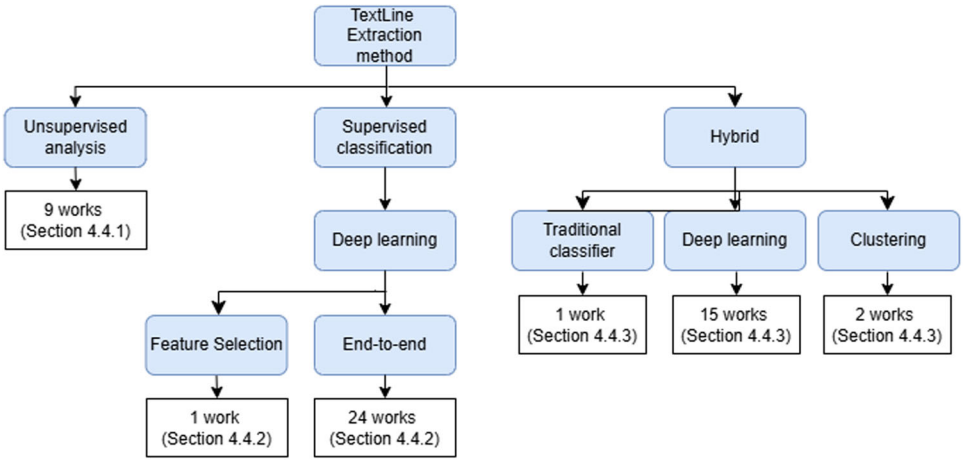


Fig. 7 Hierarchy of methods used in the surveyed publications. Some studies employed multiple approaches and were counted in several categories accordingly

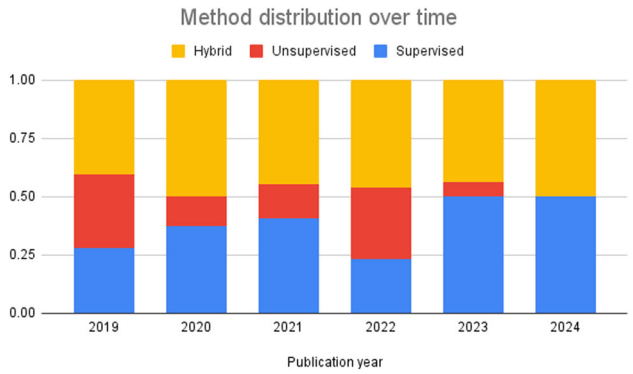


Fig. 8 Proportional methods distribution. Only one unsupervised method was introduced in 2023, and none in 2024. Hybrid and supervised methods have maintained a strong and consistent presence over the past six years

ever, the rules that guide unsupervised learning are usually designed by experts. Furthermore, unsupervised methods are known for their relatively poor performance in tasks where simple functions cannot distinguish between different classes. The computational complexity of such methods depends entirely on the defined procedure. In contrast to supervised methods, unsupervised methods do not require additional memory for saving trained models. However, they can require memory for storing pre-defined rules. Their running times can be extremely inefficient, especially if an entire input must be processed in real-time.

Table 4 provides a structured summary of the unsupervised methods applied for text line extraction. In the rest of this section, we provide an overview of the unsupervised methods. Kassis and El-Sana [50] presented a fully automated learning-free method for line detection in manuscripts. The method begins by separating components that span over multiple lines; then it removes noise and small connected components such as diacritics. The authors apply a distance transform on the image to create the image skeleton. The skeleton is pruned, and its vertexes and edges are detected to generate the initial document graph. The authors operate the graph's vertices and edges so that each edge in the final graph represents one extracted line. The authors applied the method on the DIVA-HisDB dataset on both public and private sections and have achieved results surpassing the vast majority of the competing systems.

Li et al. [61] proposed a novel two-step method for text line segmentation in Tibetan historical document recognition, which includes baseline detection and text line segmentation using the baseline. In baseline detection, the upper edges of all characters in the document were obtained by a horizontal gradient operator, then an edge connectivity definition was proposed by which the upper edge set was divided into disjoint subsets. Eligible sets were selected from these subsets, and the edges in these sets were joined in turn to obtain the baseline. In text line segmentation, the document image was truncated at the baseline position, then the adhesion regions were segmented again. Each connected region in the image was assigned to its nearest baseline. All connected regions belonging to the same baseline formed a text line.

Barakat et al. [5] evaluated a multi-oriented Gaussian-based method to segment handwritten text lines in a very complicated dataset of handwritten documents with multiple oriented and curved text lines. These text lines were written as remarks on the page margins by different writers over the years. They appear at different locations within the orientations that range between 0 and 180, or as curvilinear forms. The authors obtained a mean pixel Intersection over Union score, which is compared with the results of a single-oriented Gaussian-based text line segmentation method.

Dyla and Morain-Nicolier [26] presented a new text line segmentation and binarization method based on fast and adaptive bidimensional empirical mode decomposition. The authors apply this method to decompose each image into two components. The binarization threshold is calculated, taking into account the characteristics of each component. The inspiration for the new binarization method was drawn from the standard Niblack's and Sauvola's algorithms. Then, the text line segmentation is performed after eliminating the bidimensional intrinsic mode functions, which could disrupt the segmentation process. The results show that the proposed approach is less sensitive to noise and provides improved text line segmentation and binarization in comparison to iterative methods.

Barakat et al. [6] presented a learning-free method for text line segmentation of historical handwritten document images. This method relies on automatic scale selection together with the second derivative of anisotropic Gaussian filters to detect the blob lines that strike through the text lines. Detected blob lines guide an energy minimization procedure to extract the text lines. Historical handwritten documents contain noise, heterogeneous text line heights, skews, and touching characters among text lines. Automatic scale selection allows for automatic adaption to the heterogeneous nature of handwritten text lines in case the character height range is correctly estimated. In the extraction phase, the method can accurately split the touching characters among the text lines.

Sathik and Ratheash [87] proposed a new text line segmentation algorithm named as Text Line Slicing algorithm for Tamil palm leaf manuscripts. This article explored text line segmentation from the beginning of preprocessing. The identification, segmentation of touching, and overlapping text lines by an elongation of the character proves the uniqueness of the algorithm. Text Line Slicing provides successful results in Tamil text line segmentation amidst several challenges.

Table 4 Summary of unsupervised text line extraction methods (2019–2024)

Study	Techniques/model	Dataset(s)	Key features
Kassis and El-Sana [50], 2019	Graph-based heuristic, Skeleton pruning & distance transform	DIVA-HisDB	No training needed, interpretable graph representation. Outperformed most competing systems on DIVA-HisDB
Li et al. [61], 2019	Edge detection + baseline construction using edge connectivity	Tibetan historical documents	Effective baseline detection
Barakat et al. [5], 2019	Multi-oriented Gaussian-based method	VML-MOC	Handles curved, multi-oriented margin annotations. Improved IoU over single-oriented Gaussian method
Dyla and Morain-Nicolier [26], 2019	Bidimensional Empirical Mode Decomposition	Not specified	Noise robustness, improved segmentation and binarization. Better than iterative methods for segmentation and binarization
Barakat et al. [6], 2020	Blob line detection + energy minimization using anisotropic Gaussian filters	Diva-HisDB, READ-BAD	Handles heterogeneous text height, touching characters. Accurate line extraction, adaptable to character height
Sathik and Ratheash [87], 2021	Text line slicing based on elongation analysis	Tamil palm leaf manuscripts	Effective on overlapping lines, unique for Tamil script. Successful segmentation in challenging palm-leaf manuscripts
Nguyen et al. [72], 2022	Seam carving with novel cost functions	DIVA-HisDB, ChamDoc	High accuracy, efficient line splitting
Mukherjee et al. [70], 2022	Statistical gap analysis-based segmentation	ISIDDI, IAM-Hist, IAM	The segmentation is treated as a two-class classification problem, unsupervised parameter estimation. Outperformed ICDAR-2013 contest winner
Droby et al. [23], 2022	Global pattern differentiation, clustering-based blob lines guide seam carving	VML-AHTE	Mimics human visual perception of text lines. The combination of blob line generation and energy minimization enables effective text line segmentation without supervision

Nguyen et al. [72] proposed an effective approach for text line segmentation by analyzing the properties of document layouts. The approach combines the idea of the seam carving method with novel cost functions, to accurately split text lines. Experiments were conducted on two challenging datasets of historical documents, namely the DIVA-HisDB dataset and the ChamDoc dataset. The methods provided good results on the DIVA-HisDB dataset. On the ChamDoc dataset, the proposed method outperformed the two baseline approaches—seam carving-based and A* path planning—by a large margin.

Mukherjee et al. [70] proposed an unsupervised, robust, and simple statistical method to segment a document image that is either handwritten or machine-printed (degraded or otherwise). In this method, the segmentation is treated as a two-class classification problem. The classification is done by considering the distribution of gap size (between lines and between words) in a binary page image. The proposed method is unsupervised in the sense that no annotated document page images are necessary. Given a document page image, the parameters that are needed for the segmentation of text lines and words are learned in an unsupervised manner. The authors applied the proposed method to several popular publicly available handwritten and machine-printed datasets (ISIDDI, IAM-Hist, IAM, PBOK) of different Indian and other languages containing different fonts. The proposed method outperformed the winning method of the ICDAR-2013 handwriting segmentation contest.

Droby et al. [23] proposed an unsupervised feature learning approach for segmenting text lines of handwritten document images with no labeling effort. The authors leveraged the human visual perception of text lines and their ability to group local text line features to global coarse patterns as a supervising signal, by formulating feature learning as a global pattern differentiation task. The machine was trained to detect whether a patch contains a similar global text line pattern with its identity or neighbors and a different global text line pattern with its 90-degree-rotated identity or neighbors. Blob lines that cut across the text lines are created by clustering the central windows of patches based on the features that were retrieved from them. The blob lines direct a seam carving method that finds baselines in a color image and an energy minimization function that extracts text lines from a binary image.

Table 5 Summary of supervised text line segmentation methods (2019–2024)

Study	Techniques/model	Dataset(s)	Key Features
Mechi et al. [65], 2019	End-to-end deep learning. Modified U-Net with reduced complexity	cBAD 2017, Diva-HistBD, ANT	Efficient, reduced memory and processing requirements, avoids overfitting
Diem et al. [19], 2019	End-to-end deep learning. Describes Top-ranked system in the ICDAR 2019 competition	cBAD 2019	The winning system is composed of a CNN-based orientation estimation, an ARU-Net to detect baselines/separators, and an image processing-based methodology to extract the baselines in a parametrized form
Melnikov and Zagaynov [68], 2020	End-to-end deep learning. Lightweight CNN with CAM	cBAD 2019	Small parameter size, fast runtime, detects baseline and text line zones
Gader and Echi [31], 2020	End-to-end deep learning. AR2U-Net with attention and residuals	cBAD 2019	A pixel-wise classification to separate text-lines pixels from the background one. U-Net is modified in a way that yields better segmentation
Monnier and Aubry [69], 2020	End-to-end deep learning. FCN with ResNet-18 encoder	cBAD2 019	Memory-efficient, improved localization accuracy
Azran et al. [4], 2021	End-to-end deep learning. Dual CNN with U-Net, baselines labeling	Pinkas, cBAD 2017, cBAD 2019	Energy map-based segmentation with precise line extraction
Kodym and Hradiš [56], 2021	End-to-end deep learning. CNN baseline detection	cBAD 2019, PERO	Extending a CNN-based text baseline detection system by adding line height and text block boundary predictions to the model output, allowing the system to extract more comprehensive layout information
Li et al. [59], 2021	End-to-end deep learning. Deep neural network model, Line Counting formulation	VML-AHTE	Predicts per-pixel line number, robust to layout variance
Boillet et al. [12], 2021	End-to-end deep learning. U-Net-based FCN	cBAD 2017, DIVA-HisDB, Horae, Balsac	Pixel-level labeling, efficient architecture
Wodlinger and Sablatnig [98], 2021	End-to-end deep learning. U-Net with ResNet50 encoder	cBAD	Baseline tracking via recurrent CNN and segmentation
Dutta et al. [25], 2021	End-to-end deep learning. Multi-scale encoder-decoder CNN	cBAD 2017	Semantic segmentation with class-specific filtering
Barakat et al. [7], 2021	Deep learning for feature selection. Siamese CNN + energy minimization	Diva-HistDB, VML-AHTE	No manual annotation needed, patch similarity-based segmentation
Jia et al. [46], 2021	Deep learning for feature selection. CNN-based baseline primitive detection + relation network	cBAD 2017, cBAD2019	Anchor-free RPN + FPN + MLP improves link prediction for dense text lines
Boillet et al. [13], 2022	End-to-end deep learning. DocUFCN, ARU-Net, dhSegment ensemble	10 different datasets	Extensive experiments
Kiessling [52], 2022	End-to-end deep learning. Transformer encoder-decoder (CurT)	cBAD 2019	Bezier curve baseline modeling, no post-processing needed
Li et al. [60], 2023	End-to-end deep learning. Dynamic Text Detection Transformer	MTHv2, ICDAR 2019 HDRC-CHINESE, SCUT-CAB	Deformable attention encoder
Kaddas et al. [49], 2023	End-to-end deep learning. YOLOv5-OBB + PANet	GTLD	Rotated quadrilateral detection with CSP optimization

Table 5 continued

Study	Techniques/model	Dataset(s)	Key Features
Jian et al. [47], 2023	End-to-end deep learning. Dynamic Rotational Proposal Network + Iterative Attention Head with Mask R-CNN	CHDAC, MTHv2, and ICDAR 2019 HDRC CHINESE	Rotation-aware line detection with reduced parameters
Islam et al. [45], 2023	End-to-end deep learning. Mask R-CNN instance segmentation	Seven different datasets	Fully end-to-end, no pre/post-processing
Jindal and Ghosh [48], 2023	End-to-end deep learning. Faster R-CNN with ResNet50 backbone	Private dataset	Improved layout detection accuracy over R-CNN with VGG-16 backbone
Bernard et al. [9], 2023	End-to-end deep learning. U-FCN, R-CNN, Transformers	PARES	Transformer, and U-FCN better for structured layouts
Fizaine et al. [30], 2024	End-to-end deep learning. Comparison of U-Net vs Mask-RCNN	cBAD 2017, DiVA-HistDB, HOME-Alcar	Showed that Mask-RCNN is superior for historical document segmentation
Özşeker et al. [75], 2024	End-to-end deep learning. Pix2Pix GAN (U-Net + PatchGAN)	VML-AHTE, VML-MOC	Image-to-image translation of TLS mask predictions
Unter [95], 2023	End-to-end deep learning. Object detection models + U-Net/BiNet segmentation	Ancient Egyptian papyri	Evaluated various object detection architectures and resolution impact; segmentation from ink pixel masks
Agrawal et al. [1], 2024	End-to-end deep learning. DETR-style model + CNN-transformer hybrid (LineTR)	Multiscript Asian historical manuscripts	Two-stage system with scribble generation and seam-based line polygon detection; strong generalization and zero-shot performance

4.4.2 Supervised classification

Supervised learning with traditional machine learning (ML) algorithms usually requires expert knowledge for feature engineering, though in some cases, this requirement can be reduced by computing the principal components from raw pixels. However, the most well-known limitation of supervised learning is its dependence on annotated data for training. Supervised learning without annotated data is impossible, and an ML algorithm's performance is very much dependent on the amount and quality of its training data. Additionally, trained models have limited interpretability due to the complex and frequently non-linear separation functions learned from the training data. Nevertheless, it is very well known that supervised learning tends to provide much better results than unsupervised learning. The computational complexity varies from model to model, and often the training can be time-consuming. However, it is mainly the runtime of the inference phase that matters to the end user.

Table 5 summarizes supervised classification studies from 2019–2024, followed by a description of the primary methods used.

End-to-End Deep Neural Networks Today, deep learning networks have almost replaced traditional models due to their excellent performance and ability to operate without any need for feature engineering. These qualities allow them to produce desirable results without requiring any expert knowledge. However, because deep NNs are trained with a very large number of learning parameters, they consume a lot of annotated data; therefore, the accuracy of any deep model strongly depends on the amount and quality of the training data used. Another drawback of these models is the lack of interpretability. Many researchers complain that deep NNs can be seen as "black boxes," which need to be well-fed to produce sufficiently accurate answers (labels). These systems are frequently unable to explicitly explain why a proposed model succeeds or fails. In addition to their excellent general performance, deep learning networks provide a possibility for transfer learning in many domains, including gender classification. Transfer learning might make up for the dearth of data in a particular domain, when NNs come pre-trained on other data.

Deep NNs were applied in 21 works as a stand-alone end-to-end pipeline, including both feature selection and classification. Also, most of the hybrid approaches (11 of 13) applied DNNs (see Sect. 4.4.3).

Mechi et al. [65], as many other works on text line segmentation, applied a modified U-Net model. The U-net architecture operates both as an encoder and a decoder with two symmetric parts: (1) contracting—encoder or downsampling, performing feature extraction, and (2) expansive—decoder or upsampling, combining the contextual information captured from the first part. The authors modified the U-Net original architecture by using the "Conv2DTranspose" oper-

ation for the decoder phase (to keep the same resolution on both the input and output) and by applying a different number (half from the original) of filters in the contracting part. The proposed modifications resulted in a reduced number of parameters, memory requirements, processing time, numerical complexity, and the elimination of the over-fitting issues during the training compared with the original U-Net architecture.

Diem et al. [19] describe the cBAD ICDAR 2017 Competition on Baseline Detection with four participating methods. All methods used deep learning, two of which proposed a hybrid architecture with U-Nets and traditional post-processing. One team utilized GANs for baseline detection, with quite promising results. The winning system is supervised, and it is composed of a CNN-based orientation estimation, an ARU-Net to detect baselines and separators, and an image processing-based methodology to extract the baselines in a parametrized form. The orientation CNN was trained to estimate the predominant text orientation on a document page, and then the trained ARU-Net generated maps that encode the probability of the presence of the baseline and separator.

Melnikov and Zagaynov [68] applied a lightweight CNN with only 22K trainable parameters. The model consists of two parts: a downsampling module and a context aggregation module (CAM). The CAM module uses stacked dilated convolutions to provide large receptive field for output features. The downsampling module allows a further reduction of latency. The proposed network has two output channels, one for baseline prediction and the other for the detection of text lines without ascenders/descenders.

Gader and Echi [31] proposed AR2U-Net: a recurrent residual CNN based on the U-Net model. The authors extended U-Net with an attention mechanism, a residual structure, and recurrent convolutional layers. The AR2U-Net model was applied to allow a pixel-wise classification and to separate text-lines pixels from those in the background.

Monnier and Aubry [69] performed line-level page segmentation using a Fully Convolutional Network (FCN), optimized with a standard cross-entropy loss and followed by simple post-processing. Similarly to the work of Oliveira et al. [74], the authors adopted a simple encoder-decoder architecture, combining the descriptive power of ResNet [41] with the localization recovering capacity of U-Net [84]. In contrast to Oliveira et al. [74], the authors used a smaller ResNet-18 as a backbone encoder for saving memory while keeping document images as large as possible and also performed small modifications in the architecture for better performances. A post-processing step filtered out connected components with low area using a class-specific ratio threshold.

Azran et al. [4] applied two concatenated CNNs to preprocess the image of the document to generate an energy map, which is used to extract text baseline as a collection of minimal cost sub-seams. The first CNN is based on the U-Net and used for classifying pixels as belonging to the foreground, representing the text, or the background, keeping both the input and the output images of the same size. The second network works on fixed-size stripes to label the baseline pixels.

Kodym and Hradiš [56] extended a CNN-based text baseline detection system by adding line height and text block boundary predictions to the model output and, this way, allowed the system to extract more comprehensive layout information.

Li et al. [59] introduced the Line Counting formulation that involves counting the number of text lines from the top at every pixel location and proposed a deep NN model to perform handwritten text line segmentation based on it. The proposed model—LineCounter—is a variant of the classic encoder-decoder network with three main modules: encoder, counter, and decoder. It takes an input image, extracts convolutional features with the encoder, propagates line information along the horizontal and vertical directions in the counter, and, finally, predicts the line number for each pixel location with the decoder.

Boillet et al. [12] introduced the Doc-UFCN - a model based on U-Net, that considers the line segmentation task as a pixel-wise classification task and outputs a pixel-labeling of the input images. To design Doc-UFCN, authors chose to use the core of Yang's network [99] since it has a reduced number of parameters and contains no pre-trained parts. Therefore, the architecture of Doc-UFCN is also an FCN composed of an encoder, followed by a decoder and a final convolution layer. To keep a light model, the second decoder used in Yang's network is absent in the Doc-UFCN architecture. The researchers chose to use standard convolutions followed by transposed convolutions for the upsampling step, similar to the adaptive U-Net architecture for text line segmentation proposed in Mechi et al. [65]. This allows to have the same resolution on both sides of the network.

Wodlinger and Sablatnig [98] proposed a two-stage system for the automatic detection of text baselines of handwritten text. In the first step, a pixel-wise segmentation of the document image is performed to classify pixels as baselines, start points, end points, and backgrounds. This segmentation is then used to extract the start points of lines. Starting from these points the baseline is extracted using a recurrent CNN that directly outputs the baseline coordinates. The applied CNN is a modification of the U-Net, where the encoder is replaced with ResNet50, pretrained on ImageNet.

Dutta et al. [25] used the concept of semantic segmentation—pixel-wise labeling—with the help of a multi-scale encoder-decoder-based CNN without densely connected layers.

Barakat et al. [7] use a Siamese convolutional network, based on AlexNet, to predict whether two given document image patches are similar or different, driven by the num-

ber of foreground pixels in the patches. After the training phase, a single branch of the trained network is used to extract features of document image patches, which are in turn visualized as pseudo-RGB images and thresholded into blob lines that strike through text lines. Finally, an energy minimization framework is applied to extract the pixel labels of text lines with the assistance of the detected blob lines. The authors classify their approach as unsupervised because no manually annotated data on the text line level is needed for training. However, we put this work in the supervised category as it applies a model trained on labeled data, even if the labels (similar/different) are produced automatically.

Boillet et al. [13] presented a study conducted using three state-of-the-art systems, Doc-UFCN, dhSegment, and ARU-Net, and showed that it is possible to build generic models trained on a wide variety of historical document datasets that can correctly segment diverse unseen pages. Doc-UFCN is a U-shaped fully convolutional network. Its encoder represents the input image by 256 feature maps, and consists of dilated convolutions followed by max pooling layers. The decoder reconstructs the input image with pixel-wise labeling and contains blocks of one standard convolution followed by one transposed convolution. Finally, the features computed during the encoding step are concatenated with those of the decoding stage. The dhSegment's architecture is deeper than the Doc-UFCN one. Its encoder is composed of convolution and pooling layers (up to 2048 feature maps) and has been pre-trained on natural scene images. ARU-Net is an extended version of the standard U-Net. Two concepts have been added: spatial attention and residual structure. The spatial attention (A-Net) is a multilayer CNN that is used to handle various font sizes on a single page. The residual blocks are introduced to enable error backpropagation and identity propagation. The authors also highlighted the importance of the annotations used during training, presented a unification of the annotations, and showed its positive impact on the final text recognition results.

Kiessling [52] proposed the curve transformer (CurT), a method of direct baseline detection that models document text line detection as a set prediction of cubic Bezier curves. The proposed method simplifies the layout analysis pipeline by removing handcrafted postprocessing that is usually necessary with older algorithms. CurT has a simple Transformer-based encoder-decoder architecture and enables the processing of material ill-suited for the prevailing methods of adapting semantic segmentation backbones.

Li et al. [60] presented an end-to-end text detection framework-Dynamic Text Detection Transformer-for dense text detection in historical documents. The authors introduced a deformable convolution-based dynamic encoder to strengthen the text representation ability. Also, they designed the parallel dynamic attention heads to facilitate better interaction between the box and mask branches. Furthermore, they utilized the spatial attention transform mask head to suppress background noise in the feature maps and discrete cosine transform to encode the text masks as compact vectors for accurate text representation.

Kaddas et al. [49] applied deep NN with modifications for training. Based on YOLOv5-OBB: CSP-Darknet53 as the backbone feature extractor, a variant of Spatial Pyramid Pooling (SPP) layer feeds a Path Aggregation Network (PANet). The head of the network is composed of three convolution layers combined with the Circular Smooth Label technique to classify the quadrilateral's angle and to accurately detect the text lines in the document. In BottleNeckCSP convolutional layers, the Cross Stage Partial (CSP) technique has been incorporated to overcome the problem of redundant gradients.

Jian et al. [47] proposed a Dynamic Rotational Proposal Network (DRPN) and an Iterative Attention Head (IAH), which are incorporated into Mask R-CNN to detect text lines in historical documents. The DRPN can dynamically generate horizontal or rotational proposals to deal with multi-oriented text lines and alleviate the multi-scale problem in historical documents. The proposed IAH integrates a multi-dimensional attention mechanism for better learning of features of dense historical document text lines while improving detection accuracy and reducing the model parameters via an iterative mechanism.

In the study of Islam et al. [45], lines are treated as objects to leverage the power of deep learning-based object detection and segmentation frameworks. A deep instance segmentation model—Mask R-CNN—was trained in end-to-end fashion without any dataset-specific pre- or post-processing.

Jindal and Ghosh [48] applied a faster region-convolution NN (R-CNN). While the conventional faster R-CNN uses the VGG-16 as a backbone network to extract the spatial features, the proposed faster R-CNN structure uses Resnet50 as a backbone network. The proposed R-CNN model has outperformed the original one.

Bernard et al. [9] ran state-of-the-art algorithms—U-FCN, R-CNN, and transformers—to detect the lines within index tables on a newly released PARES dataset. Their results indicate that text line extraction works well with the U-FCN model, also indicating that transformer architectures are accurate and efficient text line detectors in historical documents.

Fizaine et al. [30] compared between Mask-RCNN- and U-Net-based networks on text segmentation of historical documents and showed the superiority of the former over the latter.

Özşeker et al. [75] reduced the TLS problem to an image-to-image translation problem. The proposed model was trained to generate the optimal mask for each text line. Image-to-image translation was utilized to map the spatial information from the original to the mask image

domain and generate a mask for each text line. Text lines in document images with challenging segmentation characteristics were segmented using a generative adversarial network (GAN). The researchers applied the pix2pix model, which is composed of two competitive CNN models, U-Net (the generator part) and PatchGAN (the discriminator part), playing a min-max game. The model was trained to learn the spatial relationship between the original manuscript texts and the mask images of the text lines.

Unter [95] explored various deep learning architectures initially designed for object detection to assess their effectiveness in identifying ancient Egyptian text lines in RGB images. It presents a comparative analysis of auxiliary training techniques, including transfer learning and data augmentation, along with an evaluation of the impact of data input resolution. Furthermore, the researchers proposed an alternative method that extracts text lines from previously binarized ink pixels, which is described in Sect. 4.4.3.

Agrawal et al. [1] introduces a two-stage text line segmentation approach designed for historical manuscripts with diverse scripts and challenging layouts. In Stage 1, input images are divided into context-adaptive patches, which are processed using a DETR-style network to generate parametric representations of text strike-through lines (scribbles) and a hybrid CNN-transformer network to create a text energy map. These outputs undergo dataset-agnostic postprocessing to produce a global scribble map. In Stage 2, the scribble map and text energy map are utilized within a seam generation framework to extract precise polygons that enclose the manuscript's text lines. Unlike prior methods, LineTR avoids dataset-specific fine-tuning, enhances zero-shot performance, and generalizes well across different document types by eliminating fixed-size patch constraints.

Deep Neural Networks for Feature Selection

The deep NN was applied as a feature extractor in the work of Jia et al. [46]. The authors introduced the concept of a baseline primitive, which is defined as a virtual bounding box centered at each baseline pixel. The proposed approach is composed of three steps: baseline primitive detection, relation learning, and baseline extraction. Baseline primitive candidates are predicted with an Anchor-Free RPN method (CNN-based) [101] and the finest feature map generated by a Feature Pyramid Network backbone [63], which is built on the top of ResNet-50. Then, a relation network—basically, a combination of CNN and MLP—is used to predict a link relationship for each pair of selected primitives. Consequently, text baselines are generated by detecting baseline primitives and grouping them with the corresponding link relationships. The design of baseline primitives aimed at extending context information to improve link prediction accuracy. The authors showed that the introduced approach can effectively detect text baselines with small inter-line or large inter-word spacing.

4.4.3 Hybrid methods

Some works chose to combine the advantages of both approaches-the precision of supervised methods and the resource efficiency of unsupervised methods. Usually, these works apply several stages of document processing, including normalization, feature extraction, grouping/generation/classification, and postprocessing for the final text line segmentation. Table 6 summarizes 16 works that introduced hybrid methods explored in our survey. We describe them in detail below. All works combined supervised and unsupervised stages.

Guerry et al. [40] proposed a system that participated in cBAD competition [19]. The system combines supervised deep learning to predict baselines, begin/end indicators of each baseline, and page boundaries; and an unsupervised rule-based method for baseline extraction.

Kiessling et al. [54] proposed a method that consists of two main stages: a pixel-level classification of baselines and a following lightweight baseline extraction step. In the first stage, a fully convolutional encoder-decoder NN assigns each pixel to a background or a baseline. The second stage is script- and layout-agnostic postprocessing operating on the heatmap produced by the NN. Baselines are vectorized into polylines, which are then used to extract a rectified rectangular line image suitable for processing by an HTR line recognition system.

Alberti et al. [3] introduced an approach integrating classification with a deep-learning model and state-of-the-art segmentation methods. The proposed method uses semantic segmentation at the pixel level as an intermediate task, followed by a text-line extraction step. The researchers demonstrated that semantic pixel segmentation can be used as a strong denoising pre-processing step before performing text line extraction.

Grüning et al. [38] presented a two-stage text line detection method for historical documents. Each detected text line is represented by its baseline. In the first stage, a deep NN called ARU-Net labels pixels to belong to one of the three classes: baseline, separator, and other. The separator class marks the beginning and the end of each text line. The ARU-Net is trainable from scratch with a few manually annotated example images. By utilizing data augmentation strategies, the researchers reduced the number of train samples to less than 50. The network predictions are then used as input for the second stage—a bottom-up clustering to build baselines.

In the study of Pastor [76], a baseline detection is reformulated as a clustering problem over a set of interest points. Its design aim is to be fast and to resist the noise artifacts that usually appear in historical manuscripts. The polyline formed

Table 6 Summary of hybrid text line segmentation methods (2019–2024)

Study	Techniques/model	Dataset(s)	Key Features
Guerry et al. [40], 2019	Supervised deep learning and an unsupervised rule-based method	cBAD 2017	The combination of heterogeneous knowledge improves baseline detection in handwritten documents. The method obtained better scores than the best method that participated in cBAD competition before 2019
Kiessling et al. [54], 2019	A fully convolutional encoder-decoder NN and postprocessing of the heatmap produced by the NN	BADAM	Presents a new dataset consisting of 400 annotated page scans of Arabic and Persian manuscripts spanning a wide range of topics and dates of production
Alberti et al. [3], 2019	A deep-learning classification model and state-of-the-art segmentation methods	DIVA-HisDB	Semantic pixel segmentation is a strong denoising pre-processing step before text line extraction
Grüning et al. [38], 2019	ARU-Net and a bottom-up clustering	Bozen	By utilizing data augmentation strategies, the number of train samples for ARU-Net is reduced to less than 50
Pastor [76], 2019	DBSCAN clustering and Extremely Randomized Trees classification	Hattem, cBAD 2017, Alcaraz	Its design aims to be fast and to resist noise artifacts
Hu et al. [43], 2021	Connected components, horizontal projection analysis, straight-line detection; CNN for character adhesion correction and stroke segmentation; stroke assignment and postprocessing	No official name	Can effectively reduce the influence of text line distortion and skew on text line segmentation, has a high degree of robustness, and has good segmentation accuracy for image text lines in Tibetan documents with touching and broken strokes
Mechi et al. [66], 2021	U-Net and modified RLSA with topological structural analysis	cBAD 2017, ICFHR18 RASM2018, ANT	The proposed method outperforms solutions presented in the RASM 2018 contest
Mechi et al. [67], 2021	A deep FCN architecture and a topological structure analysis	ANT, DIVA-HisDB, cBAD 2017, ICFHR18 RASM2018	The postprocessing refines the obtained FCN results and provides sufficient information for text recognition
Shivakumara et al. [88], 2022	Stroke width information estimation, filtering of noise, and morphological operations; DenseNet; evaluation of geometrical spatial features, the direction between candidate components and the overall text line, and the common overlapping region between adjacent components	ICDAR2019 HDRC-Chinese, ICDAR2013	The first work for segmenting text lines from handwritten documents with struck-out text. Experimental results show that the proposed method performs better for text component detection, struck-out component classification, and text line segmentation than the most representative state-of-the-art methods
Vadlamudi et al. [96], 2023	A supervised multi-task transformer deep network; an unsupervised, a scribble-conditioned seam generator; a novel diacritic feature map	ICFHR18 Asian Palm Leaf, Indiscapes2, KgathaM	A novel diacritic feature map enables improved diacritic and text line associations
Paulus et al. [77], 2023	Multiple Convolutional Autoencoder and a heuristic Seam Carving framework	ICFHR18 Asian Palm Leaf	Outperforms other handcrafted-based baselines; An adaptive slice and triple smoothing approach contribute to solving the text length variation problem; Post-processing steps are effective in connecting discontinuous medial seam
Fischer et al. [29], 2023	U-Net, a rule-based approach, DBSCAN, and Mask R-CNN	CamStd, CamMarg, GW5051, GW5056, CamManual	The main text with standard layouts was recognized in the correct reading order with a precision and recall of up to 99.9%, and complex layouts were recognized at a rate as high as 90% by using background knowledge

Table 6 continued

Study	Techniques/model	Dataset(s)	Key Features
Hu et al. [44], 2024	segmentation-based models like DBNet++, using ResNet-50 and Feature Pyramid Networks, and rule-based heuristics	CHDAC, MTHv2, ICDAR2019 HDRC Chinese	Achieves state-of-the-art on the CHDAC, MTHv2, and competitive results on ICDAR2019 HDRC Chinese datasets, with a significant improvement of 1.19% on the most challenging CHDAC dataset. Also, the method demonstrates its rotational robustness
Unter [95], 2024	A variant of U-Net called BiNet	No official name	This method derives text lines from connected components, allowing for greater control over parameters and improved identification of small ink remnants. Also, it offers increased control to a user
Gao et al. [32], 2024	Segmentation with deep NN and structural knowledge, recognition with an abductive matching mechanism and a dynamic programming algorithm; a joint optimization strategy for both models	MTHv2, GBACHD	Improves accuracy even with limited labeled data
Kundu and Bhattacharya [57], 2024	deep learning NN for object detection (YOLOv5x) with graph traversal (A* algorithm) and post-processing	ISIDDI, ISIDDI2	The proposed approach has overcome various existing challenges of line segmentation of old degraded quality documents and improved results on several publicly available datasets

by joining each couple of points in a cluster is considered a baseline. The candidate points for belonging to baselines are the local minima points of the image text edges. Some points belong to descenders, noise spots, borders, etc. Therefore, a classification system has been applied to detect points that belong to baselines. A forest of Extremely Randomized Trees [36] has been used as a classifier for minima points. Excluding the interest points classified as not belonging to the baselines, a top-down clustering algorithm based on a modified Dbscan (density-based spatial clustering of applications with noise) [27] is used to get baselines.

In the work of Hu et al. [43], a text line segmentation method based on local baselines and connected component allocation was proposed. First, the pseudo-text line is detected by analyzing the horizontal projection, straight line detection, and the average character height information, and then the local baseline position is determined. Second, adhesion area detection is performed. The position relationship between the connected components is used to complete the adhesion between characters. A CNN is used to complete the adhesion between characters and strokes. Then, touching connected components are segmented. Finally, the broken strokes are assigned to the text lines to which they belong according to the characteristics of Tibetan character structures. Subsequently, the assigned strokes are postprocessed to complete stroke correction.

Mechi et al. [66] proposed a method combining a deep network with classical document image analysis techniques for text line localization in ancient handwritten Arabic document images. The proposed method applies the U-Net model to extract the main area covering the text core. Then, a modified RLSA combined with topological structural analysis is applied to localize whole text lines.

The study of Mechi et al. [67] is based on the previous work [66] of the same authors. The paper presented a complete framework for text line segmentation in historical Arabic or Latin document images. A two-step procedure is described. First, a deep FCN architecture has been applied to extract the main area covering the text core. Researchers selected the highest-performing FCN architecture through thorough performance benchmarking of the most recent FCN architectures. Then, a post-processing step, based on topological structure analysis, is applied to extract complete text lines. This second step refines the obtained FCN results and provides sufficient information for text recognition.

Shivakumara et al. [88] proposed the text line segmentation approach applicable in the presence of both cleanly written and struck-out text. The approach consists of three steps. In the first step, components—at the word level—are detected in the input handwritten document images based on stroke width information estimation, filtering of noise, and morphological operations. In the second step, the struck-out components are identified using the DenseNet deep learning model and treated differently to clean text in further analysis. In the third step, geometrical spatial features, the direction between candidate components and the overall text line, and the common overlapping region between adjacent components are evaluated to form text lines progressively.

Vadlamudi et al. [96] introduced SeamFormer for high-precision text line segmentation. In the first stage, a supervised multi-task transformer deep network outputs coarse line identifiers (aka "scribbles") and the binarized manuscript

image. In the second stage, unsupervised, a scribble-conditioned seam generator utilizes outputs from the stage and feature maps derived from the manuscript image to produce tight-fitting line segmentation polygons. A novel diacritic feature map is incorporated, which enables improved diacritic and text line associations.

Paulus et al. [77] presented a hybrid technique that combines a convolutional-based denoising task and a heuristic Seam Carving framework. The authors proposed several changes to the original Seam Carving by application of adaptive slice, triple smoothing, and five post-processing steps aimed at reconstructing a more precise medial seam. Experimental results showed that the proposed method outperforms other handcrafted-based baselines on three palm leaf manuscript data sets. An adaptive slice and triple smoothing approach contribute to solve the text length variation problem. Finally, post-processing steps were effective in connecting discontinuous medial seam.

Fischer et al. [29] introduced a method combining the supervised and heuristic stages. First, a baseline detection based on the U-Net [65] architecture is applied. Then, a connection of similar baselines with a graph structure and computing line polygons from the detected baseline using a rule-based approach and the DBSCAN algorithm with a custom distance matrix is performed. Also, a supervised Mask R-CNN [42] is applied to find dropped capitals.

Hu et al. [44] proposed SegHist—a general framework for text line detection in Chinese historical documents. SegHist addresses challenges like high aspect ratio text lines, dense texts, and arbitrary shapes by introducing three key components: Text Kernel Stretching for improved segmentation map generation, Layout Enhanced Module to capture global layout structure, and Iterative Expansion Distance Post-processor for accurate text region recovery. SegHist is integrated with segmentation-based models like DBNet++, which use deep learning architectures like ResNet-50 and Feature Pyramid Networks (FPNs) for feature extraction, to improve detection performance. SegHist follows a supervised learning paradigm, relying on labeled segmentation maps during training. However, its post-processing method iteratively refines detected text regions, incorporating rule-based heuristics rather than purely learned parameters. This combination of learned models with handcrafted rules makes it a hybrid approach, blending supervised deep learning with algorithmic post-processing techniques for enhanced accuracy.

Unter [95] proposed an alternative method that extracts text lines from previously binarized ink pixels, which are subsequently grouped into connected components and further assembled into text lines based on their spatial relationships. The proposed method constructs the text lines from the visible ink in the image. Prior segmentation of the image is required to identify the pixels that belong to the ink. Pre-trained segmentation networks, such as U-Nets, can be used for this purpose. In [95], a variant of U-Net called BiNet [17] was used, which was specifically trained to identify the black and red ink of ancient Egyptian papyri. The segmentation result is a binary mask indicating the pixels that most likely belong to the written text.

Gao et al. [32] proposes KESAR, a novel Knowledge-Enhanced Document Segmentation and Recognition method for historical document images, built on the Abductive Learning framework. It enhances segmentation and recognition by incorporating background knowledge for character extraction and prediction, followed by joint optimization of both models. The segmentation model is trained with structural knowledge, refining character regions through abduction, while the recognition model addresses label scarcity using an abductive matching mechanism with a dynamic programming algorithm. Finally, a joint optimization strategy allows both models to improve each other's performance, aided by the Over-Segmentation and Recombination algorithm, which enhances segmentation using the character differentiation capabilities of the recognition model. KESAR is a hybrid approach that combines deep neural networks for feature extraction with symbolic reasoning to enhance segmentation and recognition, improving accuracy even with limited labeled data.

Kundu and Bhattacharya [57] proposed a hybrid approach for robust text line segmentation in degraded document images. It combines deep learning-based object detection (YOLO) with graph traversal (A* pathfinding) to enhance segmentation accuracy. The method first uses a YOLOv5x model, trained on annotated datasets, to detect initial bounding boxes for text lines. Due to the limitations of YOLO in handling skewed, overlapping, or faint text, the approach incorporates an adaptive A* algorithm to refine segmentation. Additionally, the method applies post-processing techniques to correct segmentation errors caused by asymmetric skew.

4.5 Performance evaluation

An appropriate metric is very important in evaluating and comparing the results of different algorithms. The purpose of this section is to summarize the current state-of-the-art for the evaluation of baseline detection and text line segmentation algorithms. We found that the competitions organized by the ICDAR and ICFHR conferences [18, 19, 33–35, 90, 91] contributed to the development of evaluation tools that are widely used today.

4.5.1 Baseline detection evaluation

The vast majority of the studies that addressed baseline detection utilized the evaluation scheme introduced in conjunction

with the ICDAR2017 Competition on Baseline Detection (cBAD) [18, 37]. In the rest of the paper, we will refer to this scheme as cBAD-eval.

The cBAD-eval evaluation scheme employs a coverage function that aligns hypothesis baselines with ground truth baselines and measures performance by means of Recall (R), Precision (P), and F-measure (F). The R value indicates the number of ground truth lines with corresponding hypothesis baselines, the P value penalizes segmentation errors, and the F value is the harmonic mean of the P and R:

$$F = \frac{2PR}{P+R}$$

The evaluation allows for small deviations between ground truth and hypotheses. Each page image is evaluated separately, and the average over all pages is then calculated. A detailed explanation of the evaluation procedure can be found in [37], with its corresponding implementation publicly available [39]. The cBAD-eval scheme was utilized in the ICDAR 2017 Competition on Baseline Detection [18], ICDAR2017 Competition on Layout Analysis for Challenging Medieval Manuscripts (Task 2) [90], and ICDAR 2019 Competition on Baseline Detection [19].

All but two recent studies that addressed baseline detection used cBAD-eval procedure [6, 19, 31, 38, 38, 40, 46, 52, 54, 56, 68, 69, 76, 98]. The study of Hu et al. [43] also utilized P, R, and F-measure, but used different criteria for correspondence between two baselines. The study of Li et al. [61] measured the performance by means of deviation of baseline. Table 7 summarizes the recent studies that addressed the baseline detection problem. For each study, we describe the dataset and the evaluation metric that were used.

Table 7 Summary of baseline evaluation metrics used in recent studies. For each study, we list the dataset and the evaluation metric that were used

Study	Dataset	Evaluation protocol
Kiessling [52]	cBAD:ICDAR2019 [19]	cBAD-eval
Hu et al. [43]	no official name [43]	own metric
Kodym and Hradiš [56]	cBAD:ICDAR2019 [19], PERO [56]	cBAD-eval
Wodlinger and Sablatnig [98]	cBAD:ICDAR2019 [19]	cBAD-eval
Jia et al. [46]	READ-BAD [37], cBAD:ICDAR2019 [19]	cBAD-eval
Barakat et al. [6]	Diva-HisDB [28], READ-BAD [37]	cBAD-eval
Melnikov and Zagaynov [68]	CDAR 2017 cBAD [37]	cBAD-eval
Gader and Echi [31]	BADAM [53]	cBAD-eval
Monnier and Aubry [69]	READ-BAD [37], cBAD:ICDAR2019 [19]	cBAD-eval
Kiessling et al. [54]	BADAM [53]	cBAD-eval
Li et al. [61]	no official name [61]	own metric
Grüning et al. [38]	READ-BAD [37], DIVA-HisDB [28], Bozen [93]	cBAD-eval
Pastor [76]	ALCARAZ [76], HATTEM [86]	cBAD-eval
Guerry et al. [40]	READ-BAD [37]	cBAD-eval
Diem et al. [19]	cBAD:ICDAR2019 [19]	cBAD-eval

4.5.2 Text line segmentation evaluation

For the evaluation of text line segmentation, the Intersection over Union (IU), Precision (P), Recall (F), and F-measure are widely used. The detailed analysis of recent studies revealed that the two most extensively used protocols are the Handwriting Segmentation Contest protocol and the DIVA evaluation protocol. Throughout the remainder of the paper, we refer to these protocols as HSCP and DIVA-eval, respectively, and overview them in Sects. 4.5.3 and 4.5.4, respectively. We found that DIVA-eval and HSCP are the most widely utilized metrics with 27.9% and 25.6% studies, respectively, 32.6% studies used other protocols based on P, R, and F-measure, and 14.05% utilized additional evaluation metrics (see Fig. 9). Among additional metrics that were utilized in several studies are Hausdorff Distance and its variants [1, 51], AP, mAP, threshold-free mAP [13, 30, 95], and ZoneMap metric [66, 67]. Table 8 summarizes the recent studies that addressed the text line segmentation problem. For each study, we describe the dataset and the evaluation protocol that were used.[3]

4.5.3 Handwriting segmentation contests protocol

The Handwriting Segmentation Contests Protocol (HSCP) was used in ICDAR 2007, ICDAR 2009, ICDAR 2013, and ICFHR 2010 Handwriting Segmentation Contests [33–35, 91], and ICFHR 2018 Competition On Document Image Analysis Tasks for Southeast Asian Palm Leaf Manuscripts (Task 2) [51].

This evaluation protocol is based on counting the number of matches between the text lines detected by the algorithm

[3] Several studies experimented with both historical and modern datasets. Because this survey focuses on historical documents analysis, Table 8 lists only historical datasets that were utilized in the studies.

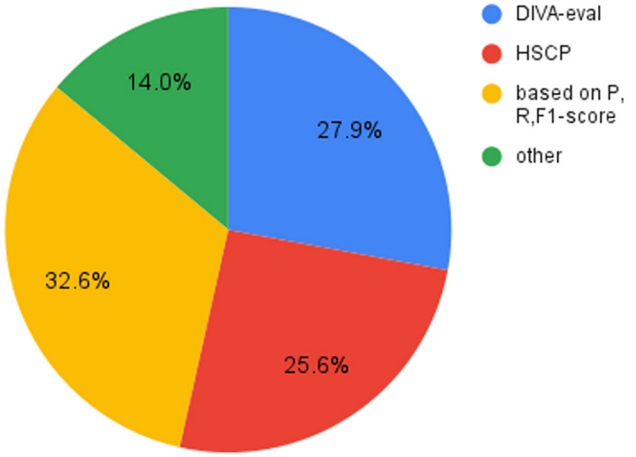


Fig. 9 Text line segmentation evaluation protocols utilized in recent studies

and the text lines in the ground truth. The evaluation assumes that both the ground truth and the result are image files with zeros representing the background and positive integer values corresponding to a segmentation region.

The output consists of three metrics: detection rate DR (similar to Recall), recognition accuracy RA (similar to Precision), and performance metric FM (Similar to F-measure).

For a given image, let R_i represent the set of ON points inside the ith detected line segment, G_j represent the set of ON points inside the jth line segment in the ground truth, and $T(p)$ be a number of points in a given set p. Then, the $MatchScore(i, j)$ measure is calculated as follows:

$$MatchScore(i, j) = \frac{T(G_j \cap R_i)}{T(G_j \cup R_i)}$$

A region pair (i, j) is considered a one-to-one match if the $MatchScore(i, j)$ is equal to or greater than the threshold T_a. Let N and M represent the numbers of ground-truth and detected lines, respectively, and let $o2o$ represent the number of one-to-one matches. The tool calculates the DR, RA, and FM as follows:

$$DR = \frac{o2o}{N}, RA = \frac{o2o}{M}, FM = \frac{2 \times DR \times RA}{DR + RA}$$

4.5.4 Diva evaluation metrics

The ICDAR2017 Competition's text line segmentation evaluation (Task 3) [90] utilized an open evaluation tool for layout analysis presented in [2], and freely available at [20]. This tool relies on the Intersection over Union (IU) metric:

$$IU = \frac{TP}{TP + FP + FNP + R}$$

where TP represents True Positives, FP represents False Positives, and FN represents False Negatives. Both Line UI and Pixel UI are calculated. The Pixel IU is based on the number of foreground pixels inside the text line.

Two polygons, the ground truth and the prediction, are considered a matching pair if both the pixel precision and recall of these two polygons exceed a predefined threshold. The ICDAR2017 Competition [90] employed a threshold of 75%. If the pixel precision falls below the threshold, it is labeled as an FP. If the recall falls below the threshold, it is labeled as an FN. Pixel UI and Line UI are calculated for each page, with the mean Line IU, mean global Pixel IU (PIU) and the mean Matched Pixel IU (MPIU) over all the pages is reported. MPIU is calculated over the different lines of each page.

4.6 Comparison of the results on benchmark datasets

This section compares and analyzes the results of studies that utilized the cBAD 2017, cBAD 2019, and DivaHisDB benchmark datasets.

4.6.1 Comparison of the results on the cBAD 2017 dataset

In this section, we compare the results of the studies that utilized the cBAD 2017 dataset. The CBAD 2017 dataset was utilized both for baseline detection and text line extraction, and the corresponding results are presented and compared in Tables 9 and 10, respectively.

An analysis of Table 9 reveals that methods based on the U-net architecture achieved top performance on Track A. The approach proposed by Grüning et al. [38] shows the best scores, closely followed by the method presented by the DMRZ group at the cBAD competition. Both U-net-based methods exhibited strong performance across all evaluated metrics. The method by Jia et al. [46] employing an Anchor-Free RPN approach achieved competitive results, ranking third. The fourth-best method also leveraged a U-net architecture [69]. The learning-free method based on energy-minimization [6] has a respectable recall of 0.911, which is competitive with some learning-based methods. This high recall indicates that the method is quite good at identifying relevant instances, but it may not be as precise, which could lead to higher false positives as reflected in its lower precision (0.821).

The increased challenges in Track B are reflected in generally lower performance metrics. The methods of [46], [69], and [38], while still leading, show a slight decrease compared to their performance in Track A.

Overall, the dominance of U-net-based approaches among the top best results in both tracks suggests that this architecture is well-suited for the baseline detection task.

Table 8 Summary of text line evaluation metrics used in recent studies. For each study, we list the historical dataset and the evaluation metric that were used

Study	Dataset	Evaluation protocol
Paulus et al. [77]	ICFHR18 Asian Palm Leaf [51]	HSCP
Li et al. [60]	MTHv2 [64], ICDAR 2019 HDRC-CHINESE [85], SCUT-CAB [15]	DIVA-eval
Kaddas et al. [49]	GTLD [49]	P, R, F-measure, mAP
Vadlamudi et al. [96]	ICFHR18 Asian Palm Leaf [51], Indiscapes2 [82], KgathaM [96]	IU, Hausdorff Distance and its variants
Jian et al. [47]	CHDAC [47], MTHv2 [64], ICDAR 2019 HDRC CHINESE [85]	P, R, F-measure
Islam et al. [45]	VML-AHTE [24], READ-BAD [37], Digital Peter [80], DIVA-HisDB [28], ICDAR2017 Batch A [94], Bozen [93]	DIVA-eval
Jindal and Ghosh [48]	no official name [48]	P, R, F-measure
Shivakumara et al. [88]	ICDAR2019 HDRC-Chinese [85]	HSCP
Mukherjee et al. [70]	ISIDDI [10], IAM-HistDB [28]	HSCP
Boillet et al. [13]	AN-Index [13], Balsac [97], BNPP [13], Bozen [93], DIVA-HisDB [28], HOME-NACR [14], HORAE [11], ScribbleLens [22]	analyzed several segmentation metrics (P, R, F-measure, mAP, AP@t), introduced mean threshold-free mAP
Nguyen et al. [72]	DivaHisDB [28], ChamDoc [72]	DIVA-eval (on DivaHisDB), HSCP (on ChamDOC)
Droby et al. [23]	VML-AHTE [24], DIVA-HisDB [28]	DIVA-eval
Azran et al. [4]	Pinkas [58]	HSCP
Mechi et al. [66]	READ-BAD [37], ICFHR18 RASM2018 [16], ANT [65]	P, R, F-measure, and ZoneMap
Mechi et al. [67]	READ-BAD [37], READ, DIVA-HisDB [28], ICFHR-RASM 2018 [16], ANT [65]	P, R, F-measure, and ZoneMap
Boillet et al. [12]	READ-BAD [37], DIVA-HisDB [28], HORAE [11], Balsac [97]	P, R, F-measure
Barakat et al. [7]	DIVA-HisDB [28], VML-AHTE [24]	HSCP, DIVA-eval
Mechi et al. [65]	READ, READ-BAD [37], Diva-HisDB [28], ANT [65]	HSCP
Alberti et al. [3]	DIVA-HisDB [28]	DIVA-eval
Barakat et al. [5]	VML-MOC [5]	DIVA-eval
Kassis and El-Sana [50]	DIVA-HisDB [28]	DIVA-eval
Hu et al. [43]	no official name [43]	HSCP
Kodym and Hradiš [56]	PERO [56]	P, R, F-measure
Barakat et al. [6]	Diva-HisDB [28], READ-BAD [37]	DIVA-eval
Li et al.[61]	NMU-THDTL [61]	P, R, F-measure
Pastor [76]	ALCARAZ [76], HATTEM [86]	HSCP
Dutta et al. [25]	READ-BAD [37]	P, R, F-measure
Bernard et al. [9]	PARES[8]	IU, P, R, F-measure
Fizaine et al. [30]	DRoSB[30], READ-BAD [37], DIVA-HisDB [28], HOME-Alcar [92]	DIVA-eval, AP, AP@[0.5,0.95], object level metrics
Fischer et al. [29]	READ-BAD [37], CamStd [29], CamMarg [29], GW5051 [29], GW5056 [29], CamManual [29]	DIVA-eval
Gao et al. [32]	MTHv2 [64], GBACHD [100]	R, P, F-measure
Özşeker et al. [75]	VML-AHTE [24], VML-MOC [5]	DIVA-eval
Hu et al. [44]	CHDAC [47], MTHv2 [64], HDRC-Chinese [85]	R, P, F-measure
Agrawal et al. [1]	ICFHR18 Asian Palm Leaf[51], Indiscapes2 [82], KgathaM [96], no official name [1]	Average Hausdorff Distance, IoU
Unter [95]	no official name [95]	P, R, IoU, mAP
Kundu and Bhattacharya [57]	ISIDDI [10]	cBAD-eval

Table 9 Comparison of the baseline detection results on cBAD2017 dataset

Paper	Year	Method	Track A			Track B		
			P	R	F	P	R	F
DMRZ (cBAD competition [18])	2017	residual U-net	0.973	0.97	0.971	0.854	0.863	0.859
UPVLC (cBAD competition [18])	2017	clustering over a set of interest points using DBScan technique	0.937	0.855	0.894	0.833	0.606	0.702
BYU (cBAD competition [18])	2017	FCN combined, followed by probabilistic Hough lines	0.878	0.907	0.892	0.773	0.82	0.796
IRISA (cBAD competition [18])	2017	blurred image combined with a description of text lines in the context of the document structure	0.883	0.887	0.88	0.692	0.772	0.73
LITIS (cBAD competition [18])	2017	FCN-based	0.78	0.836	0.807	–	–	–
Grüning et al. [38]	2019	ARU-Net (a model based on U-net)	**0.98**	**0.98**	**0.978**	**0.93**	0.92	0.922
Guerry et al. [40]	2019	combination of dhSegment CNN with structural information expressed in a syntactical way	–	–	–	0.858	0.935	0.895
Melnikov and Zagaynov [68]	2020	lightweight CNN, consisting of a downsampling and a context aggregation modules	0.943	0.956	0.949	0.888	0.923	0.905
Barakat et al. [6]	2020	energy minimization with graph cuts (learning-free)	0.821	0.911	0.863	–	–	–
Monnier and Aubry [69]	2020	FCN based on U-net	0.948	0.978	0.963	0.883	**0.947**	0.914
Jia et al. [46]	2021	Anchor-Free RPN (built on the top of ResNet-50)	0.965	0.978	0.972	0.928	**0.947**	**0.938**

The bold values indicate the best results

The cBAD 2017 dataset has been also utilized for text line extraction in the [4, 12, 13, 25, 65–67]. However, comparing results across these studies proves challenging due to the variability in experimental settings and the range of applied evaluation metrics. Table 10 provides a summary of the studies that has been utilized for text line extraction. Each row outlines the methodology applied, the evaluation metrics used, and the key reported results.

4.6.2 Comparison of the results on the cBAD 2019 dataset

The cBAD 2019 dataset was mostly utilized for baseline detection. In addition to the five teams that participated in the cBAD2019 competition, we identified five studies that used the same evaluation settings and metrics. The results are compared and summarized in Table 11.

The highest F score (0.938) is achieved by the method proposed by Jia et al. [46], which utilizes Anchor-Free RPN. The Planet team from the cBAB 2019 competition with a hybrid architecture using ARU-Nets comes in a close second (F=0.931). The third highest result with F=925 is achieved by Monnier and Aubry [69], which again is based on the U-Net architecture.

We also found two studies that utilized the cBAD 2019 dataset for the TL task [4, 13]. The study of Azran et al. [4] applied a combination of two CNNs. Boillet et al. [13] explored Doc-UFCN, dhSegment, and ARU-Net and compared them using a wide range of evaluation metrics.

4.6.3 Comparison of the results on the ICDAR 2017 (DivaHisDB) dataset

The DivaHisDB was utilized in the ICDAR2017 Competition on Layout Analysis for Challenging Medieval Manuscripts [90], both for baseline detection and text line segmentation (Tasks 2 and 3, respectively). Training, validation, and test sets consist of 30 pages, 10 pages, and 10 pages per manuscript, respectively. The ground truth for text line segmentation is provided in PAGE XML format.

Table 12 lists the results for baseline detection on DivaHisDB. The highest overall F-measure score (98.99) is achieved by the method proposed by Grüning et al. [38], which utilizes ARU-Net. The ARU-Net shows stable performance on all manuscripts. Closely following are the results of the learning-free method proposed by System-8 from the ICDAR 2017 competition, with an overall 98.22 F-measure score. This method uses a combination of Adaptive Run Length Smoothing and seam carving algorithms.

Table 13 lists the results for text line segmentation on DivaHisDB. The top two best results are achieved by the methods of Alberti et al. [3] and Nguyen et al. [72]. Both methods utilize a seam-carving algorithm. It seems that despite being a relatively old technique, seam-carving

Table 10 Comparison of the text line detection results on cBAD2017 dataset

Study	Year	Method	Metric	Reported results
Mechi et al. [65]	2019	Adaptive U-Net	HSCP	P=0.75, R=0.85, F=0.75, IU=0.65, thresh = 0.3
Mechi et al. [67]	2021	FCNs followed by topological structural analysis, with U-Net showing the best performance	P, R, F-measure, ZoneMap	Track A: P=75.9%, R=85.1%, F=79.1%, Track B: P=67.4%, R=80.5%, F=70.5%
Dutta et al. [25]	2021	A multi-scale encoder-decoder-based CNN model	P, R, F-measure	Track A only: P=99.7%, R=99.2%, F-measure = 99.49%
Boillet et al. [12]	2021	Doc-UFCN (based on U-Net)	IU, P, R, F-measure	Performed multiple experiments under different scenarios
Azran et al. [4]	2021	A combination of two variants of CNNs, followed by minimal energy seam extraction	HSCP	RA=98.95%, DR=98.96%, FM=98.95%
Mechi et al. [66]	2021	U-Net architecture to extract text lines at X-height level, followed by a combination of run length smoothing algorithm and topological structural analysis	ZoneMap	Experimented with a selected set of 337 images. Match = 6374, Miss=1106, False alarm = 865, Split = 439, Merge = 39
Boillet et al. [13]	2022	Experimented with Doc-UFCN, dhSegment, and ARU-Net	P, R, F-measure, MAP, AP@t	Analysis and comparison of the evaluation metrics
Fizaine et al. [30]	2024	Mask-RCNN	DIVA-eval, AP, AP@[0.5,0.95], object level metrics	IU = 0.64, F-measure = 0.76

Table 11 Comparison of the baseline detection results on cBAD2019 dataset

Study	Year	Method	P	R	F
Planet (cBAD 2019 competition [19])	2019	a hybrid architecture with ARU-Nets	**0.937**	0.926	0.931
DMRZ (cBAD 2019 competition [19])	2019	based on a combination of residual U-net	0.925	0.905	0.915
UPVLC (cBAD 2019 competition [19])	2019	CGANN followed by a DBScan clustering algorithm	0.911	0.902	0.907
TJNU (cBAD 2019 competition [19])	2019	based on a FCN	0.852	0.885	0.868
Baseline (cBAD 2019 competition [19])	2019	CNN for classifying basic document properties and a second that detects and extracts baselines	0.773	0.743	0.758
Monnier and Aubry [69]	2020	FCN based on U-net	0.92	0.931	0.925
Jia et al. [46]	2021	based on the Anchor-Free RPN	0.933	**0.943**	**0.938**
Kodym and Hradiš [56]	2021	CNN-based	0.906	0.897	0.902
Wodlinger and Sablatnig [98]	2021	CNN-based	0.872	0.89	0.881
Kiessling [52]	2022	Transformer-based	0.909	0.908	0.908

The bold values indicate the best results

Table 12 Comparison of F-measure for baseline detection on ICDAR 2017 dataset

Study	Year	Method	F-measure			
			CB55	CSG18	CSG863	Overall
System-8 (ICDAR2017 competition [90])	2017	Adaptive Run Length Smoothing and seam carving algorithms	98.96	98.53	97.16	98.22
System-2 (ICDAR2017 competition [90])	2017	FCN-based	95.97	**98.79**	98.30	97.68
System-7 (ICDAR2017 competition [90])	2017	Gaussian mixture models to classify connected components, followed by energy-based baseline detection	95.34	87.34	97.51	93.40
Grüning et al. [38]	2019	ARU-Net	**99.8**	98.28	**98.89**	**98.99**

The bold values indicate the best results

remains a powerful tool for extracting text lines. Additionally, seam-carving has the advantage of being learning-free and can be applied in a scenario in which data is scarce. We can see two additional learning-free methods: [6], which perform well with high PIU and LIU across all categories, and [50], which showed results comparable to some learning-based methods. Looking at overall PIU and LIU (the last two columns), methods from recent years (2020-2023) generally outperform earlier methods. This suggests advancements in layout analysis techniques.

We also identified four additional studies that performed experiments with the DivaHisDB for text line detection. These studies utilized different evaluation metrics and are not directly comparable. We summarize these studies in Table 14.

4.7 Available software resources

Today, the research community not only encourages the sharing of data, thereby creating datasets that allow researchers around the world to compare their results but also to the sharing of source code, which contributes to consistent evaluation. The alternative to publicly available code is a reproduction of the approach of interest, which is not always an easy task. Consequently, because many small details are usually not presented in a paper, an attempted reproduction of a particular research project might produce different results, causing inconsistency in the reported evaluation. Publicly available code enables its adaptation and application on new datasets, which mutually contributes to two interest groups: the method's authors, who acquire another citation, and the other publication, when more extensive experiments reach the community.

We examined the surveyed papers reporting the best-performing systems (see Tables 9 to 14) and checked whether they provided available implementations of the introduced methods. Unfortunately, only two papers included links to their methods' implementation: [38],[4] introducing ARU-Net, and [3],[5] presenting a method based on semantic segmentation with ResNet-18 and seam carving.

5 Discussions and future research directions

The goal of this study was to review and analyze recent advances in text line extraction and baseline detection for historical document images. This systematic survey was performed by answering research questions based on works published since he beginning of 2019, which were scrupulously selected by the survey authors. Below, we summarize the major findings of this systematic study. We also highlight the identified gaps and suggest future research directions.

- **Research Demographic Information**. The papers covered in our study were written by researchers from 22 different countries. France leads with the greatest number of papers, followed by China, Israel, and India.
- **Research Relevance**. Conference publications dominated the field of text line extraction and baseline detection over the past six years. However, journal publications have been observed being dominant in 2022 and almost equal to the number of conference publications in 2023. We also can observe that, despite reaching its peak in 2021, this topic is still active and quite popular.
- **Benchmark Datasets**. We identified 44 datasets that provide the annotation on text line or baseline levels. The majority of the datasets (24) are available for download,

[4] https://github.com/TobiasGruening/ARU-Net.

[5] https://github.com/DIVA-DIA/Text-Line-Segmentation-Method-for-Medieval-Manuscripts.

Table 13 Comparison of the results for text line detection on DivaHisDB dataset, DIVA-eval protocol. LIU denotes Line IU, and PIU denotes Pixel IU

Study	Year	Method	CB55		CSG18		CSG863		Overall	
			PIU	LIU	PIU	LIU	PIU	LIU	PIU	LIU
System-8 (ICDAR2017 competition [90])	2017	Adaptive Run Length Smoothing and seam carving algorithms	93.75	99.33	94.47	94.9	90.81	96.75	93.01	96.99
System-2 (ICDAR2017 competition [90])	2017	FCN-based	80.23	84.26	75.31	69.57	93.68	90.64	83.07	81.5
System-6 (ICDAR2017 competition [90])	2017	FRRN network	30.53	5.67	54.52	39.17	46.09	25.96	43.71	23.6
Kassis and El-Sana [50]	2019	Distance Transform Graph (learning-free)	–	84.88	–	82.6	–	80.02	–	82.5
Alberti et al. [3]	2019	combined semantic segmentation ResNet-18 and seam carving	–	–	–	–	–	–	97.22	**100**
Barakat et al. [6]	2020	energy minimization with graph cuts (learning -free)	97.04	100	98.14	98.38	97.31	100	97.5	99.46
Barakat et al. [7]	2021	siamese network based on AlexNet (unsupervised)	80.35	77.3	94.3	95.5	90.58	89.4	–	–
Nguyen et al. [72]	2022	based on seem carving (learning free)	–	–	–	–	–	–	**99.36**	98.86
Droby et al. [23]	2022	Siamese-based (unsupervised)	97.64	100	97.79	97.65	97.18	100	–	–
Islam et al. [45]	2023	based on Mask R-CNN	96.1	98	95.9	98.6	96	98.7	96	98.4

The bold values indicate the best results

Table 14 Comparison of the results for text line detection on DivaHisDB dataset, different evaluation protocols

Paper	Year	Method	Metric	Reported results
Mechi et al. [65]	2019	Adaptive U-Net	HSCP (P, R, F-measure with different IU)	best F results: 0.76 (CB55), 0.65 (CSG18), 0.83 (CSG863)
Mechi et al. [67]	2021	FCNs followed by topological structural analysis, with dilated FCN showing the best performance	cBAD-eval, ZoneMap	F = 0.92
Boillet et al. [12]	2021	Doc-UFCN (a model based on U-Net)	IU, P, R, F-measure	multiple experiment, the best results: mean IU=0.75, P = 0.92, R=0.8, F-measure=0.86
Boillet et al. [13]	2022	experimented with Doc-UFCN, dhSegment, and ARU-Net	analyses several segmentation metrics (P, R, F-measure, MAP, AP@t)	Analysis and comparison of the evaluation metrics
Fizaine et al. [30]	2024	Mask-RCNN	DIVA-eval, AP, AP@[0.5,0.95], object level metrics	IU = 0.55, F-measure = 0.7

and the corresponding links are listed in Table 2. The datasets cover 22 different scripts, with Latin and Arabic being the most represented. The most extensively utilized datasets are cBAD/READ-BAD, DivaHisDB, and ICDAR 2019 cBAD datasets.

However, there still exist groups of languages that are less studied (e.g., the Slavic group of languages that include Ukrainian, Polish, and Serbian, and the Baltic languages that include Lithuanian and Latvian). Future research should focus on collecting and annotating datasets for underrepresented languages. Creating benchmark datasets for these language groups will enable consistent evaluation and comparison of different models and approaches.

In addition to creating the datasets for underrepresented scripts, we advocate for open-access datasets. Dataset availability promotes transparency and reproducibility and allows validation of findings by independent researchers. Open access also encourages collaboration and further advancements in the field.

- **Methods**. As we observed during our study, the most popular models in the last six years were based on CNNs. Deep models were applied in works from both supervised and hybrid categories, which were dominant among all approaches proposed recently. Only nine out of 49 works introduced unsupervised methods. According to our observations, most of the studied works do not motivate or explain their choice of models. In the future, there is a need for more motivated approaches, explainable models, and ablation studies, revealing which stages of a document's processing affect text line detection most. This information might help researchers make the right choices for a specific type of data.
- **Evaluation Protocols and Comparison of the Results**. We found a diversity of evaluation methods used in recent studies, but it seems the majority chose to follow well-established protocols. The competitions on the text line extraction and baseline detection contributed significantly to developing evaluation protocols that have been widely accepted within the research community. For baseline extraction, the cBAD evaluation protocol is widely utilized. For text line extraction, two well-established protocols are widely used: the Handwriting Segmentation Contests Protocol and the Diva Evaluation Protocol. We encourage researchers and developers to adopt standard protocols. Utilizing established evaluation methods not only enhances the comparability of results but also can help identify and address the specific challenges and limitations of existing methods.

  As a future research direction, we suggest including an evaluation of the computational efficiency as part of the evaluation protocols. This will encourage the development of scalable methods for real-time application.
- **Available Software Resources**. We checked the papers reporting the best results for the availability of their code. Unfortunately, only two available sources were found. There is an obvious need for the available code sources for the sake of consistent comparisons in future research.

6 Conclusions

This paper aimed to systematically identify, analyze, and compare recent research on text line extraction and baseline detection in historical document images. Guided by research questions and a structured data extraction and analysis process, this review highlights several important findings. We summarized valuable statistical information about historical datasets that provide annotations at the text line level, overviewed and categorized recent methods, summarized existing evaluation protocols, and compared the results of different methods on standard benchmark datasets.

Finally, we identified gaps and suggested future research directions. Future research will benefit from the adoption of standard protocols and a focus on computational efficiency. In addition, addressing underrepresented languages by developing and sharing open-access benchmark datasets will contribute to further advancements in the field.

More theoretically and empirically supported explanations about model choices and adaptations in future research would bring important insights and recommendations for further development of the area. We recommend that researchers perform detailed ablation studies in their works, to provide a better understanding of which preprocessing and which stages at the proposed method fit a specific type of data.

Also, the available implementations of the introduced methods would be very useful for comparative experiments in future works.

Author contributions I.R. and M.R. contibuted equally to this work

Funding Open access funding provided by Sami Shamoon College of Engineering.

Data Availability No datasets were generated or analysed during the current study.

Declarations

Conflict of interest The authors have no conflict of interest to declare.

Competing interests The authors declare no competing interests.

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